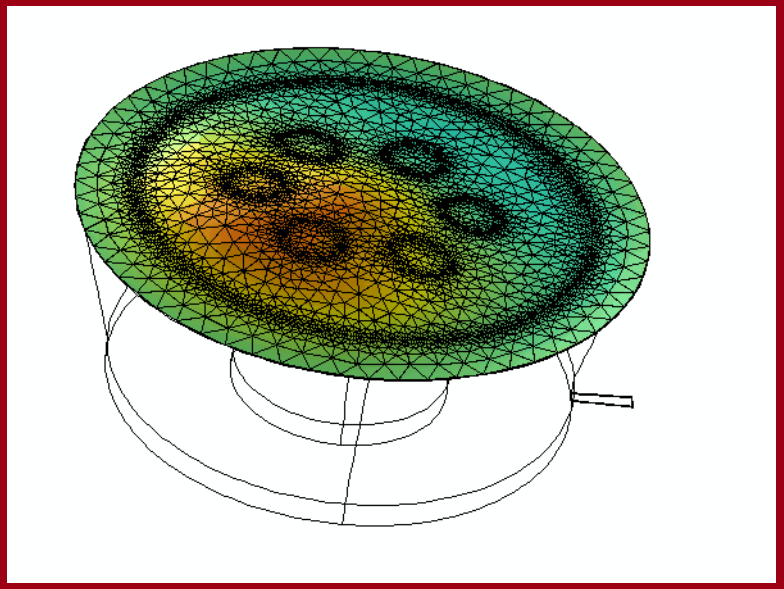
Università degli Studi di Padova

DIPARTIMENTO DI INGEGNERIA DELL'INFORMAZIONE



Digital Forensics & Biometrics

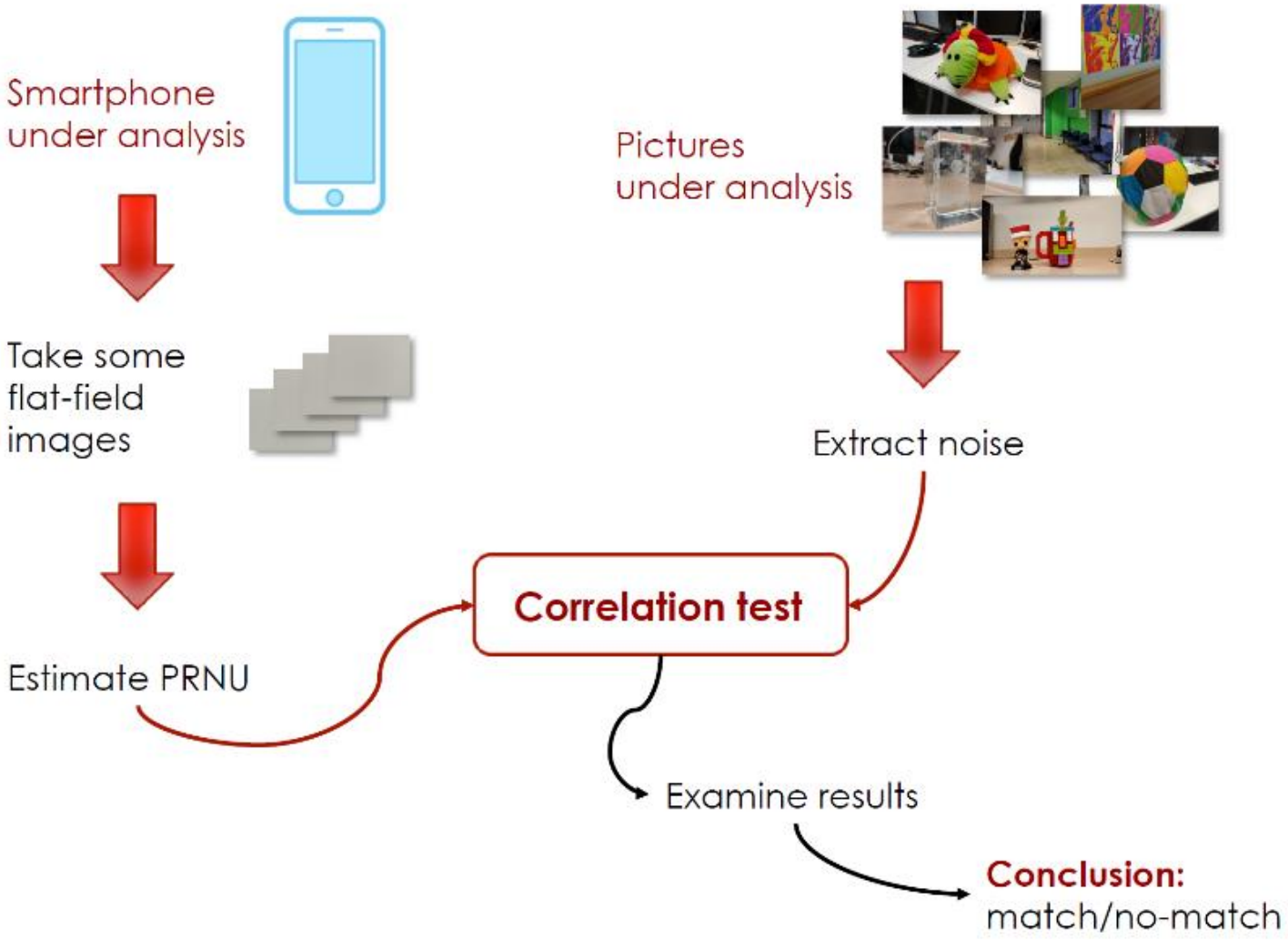
A.A. 2026/2027

Lecture 10 – camera forensics

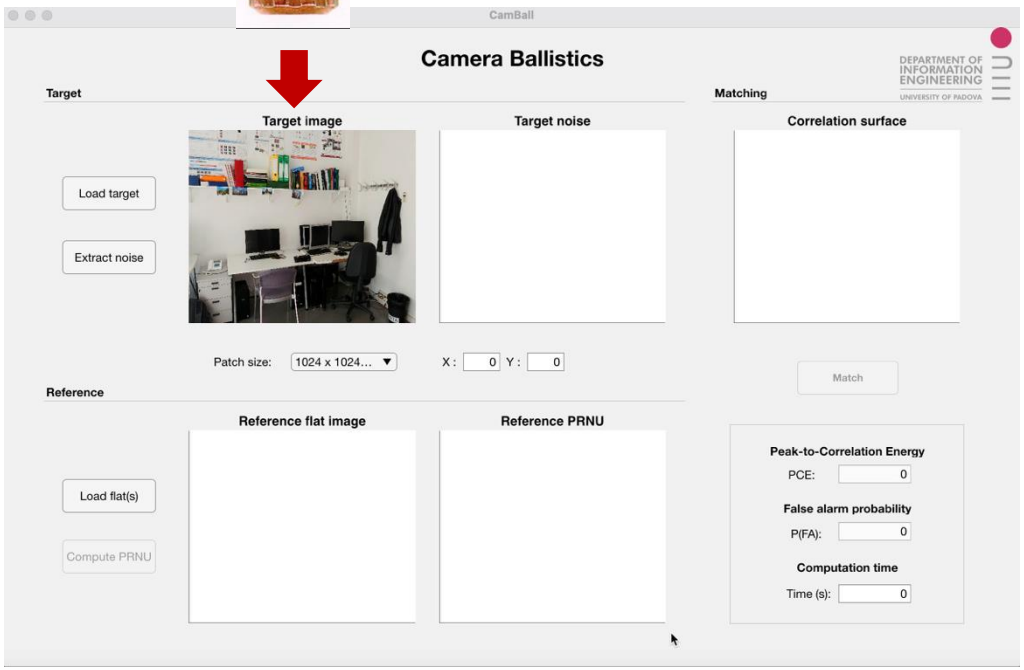
Simone Milani
Room 216
DEI A

Phone: 049 827 7641
E-mail:
simone.milani@dei.unipd.it

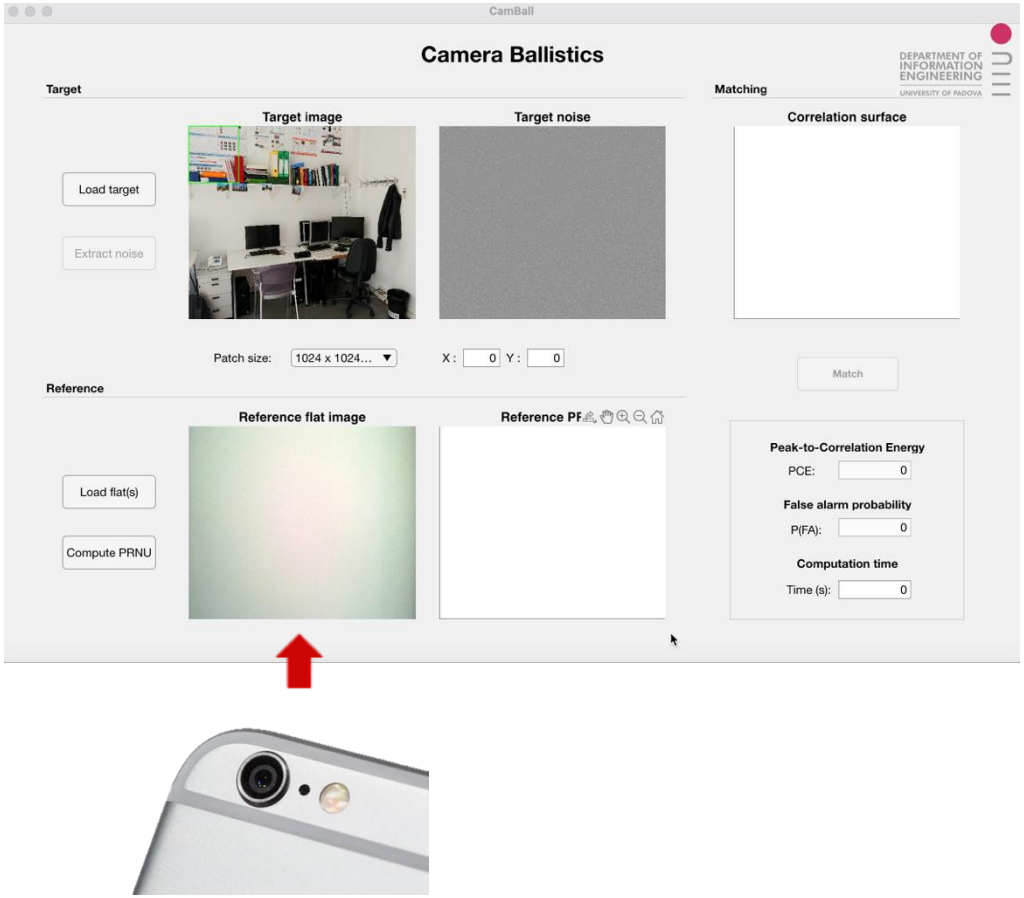
PRNU IN PRACTICE

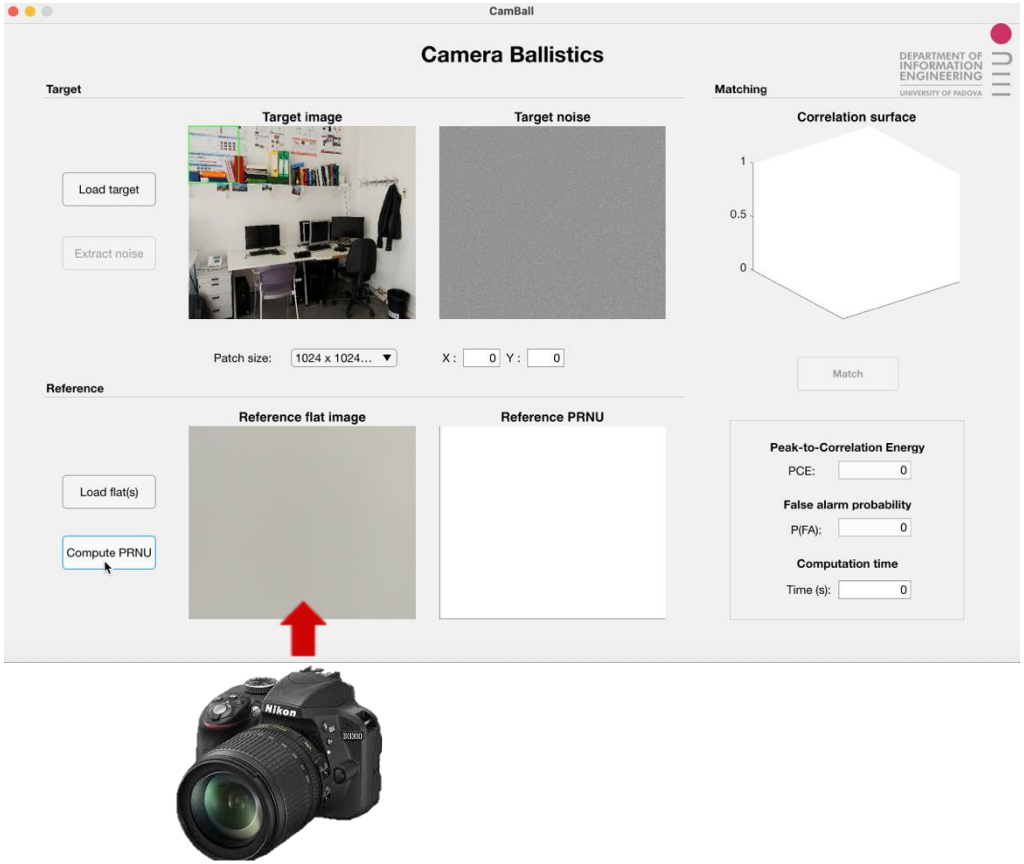





CamBall
Camera Ballistics
DEPARTMENT OF INFORMATION ENGINEERING
UNIVERSITY OF PADOVA
Target
Target image
Target noise
Matching
Correlation surface
Load target
Extract noise
Patch size:
1024 x 1024…
X : 0 Y : 0
Match
Reference
Reference flat image
Reference PRNU
Load flat(s)
Compute PRNU
Peak-to-Correlation Energy
PCE: 0
False alarm probability
P(FA): 0
Computation time
Time (s): 0

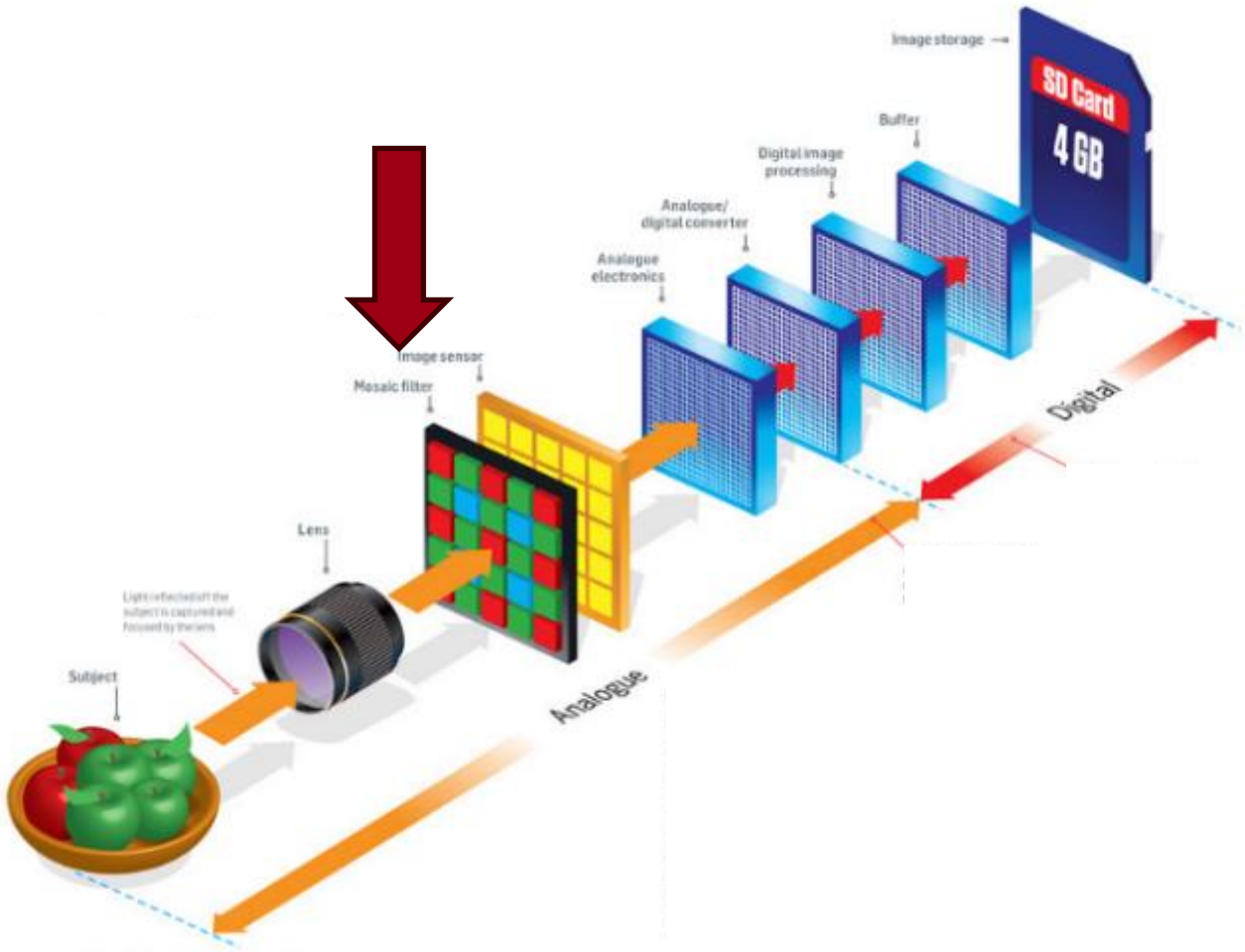



CamBall
Camera Ballistics
DEPARTMENT OF INFORMATION ENGINEERING
UNIVERSITY OF PADOVA
Target
Target image
Target noise
Load target
Extract noise
Patch size:
1024 x 1024…
X :
0
Y :
0
Reference
Reference flat image
Reference PF
Load flat(s)
Compute PRNU
Matching
Correlation surface
Match
Peak-to-Correlation Energy
PCE:
0
False alarm probability
P(FA):
0
Computation time
Time (s):
0


CamBall
Camera Ballistics
DEPARTMENT OF INFORMATION ENGINEERING
UNIVERSITY OF PADOVA
Target
Target image
Target noise
Load target
Extract noise
Patch size:
1024 x 1024...
X : 0 Y : 0
Reference
Reference flat image
Reference PRNU
Load flat(s)
Compute PRNU
Matching
Correlation surface
1
0.5
0
Match
Peak-to-Correlation Energy
PCE: 0
False alarm probability
P(FA): 0
Computation time
Time (s): 0

Nikon

DIGITAL CAMERA MODEL

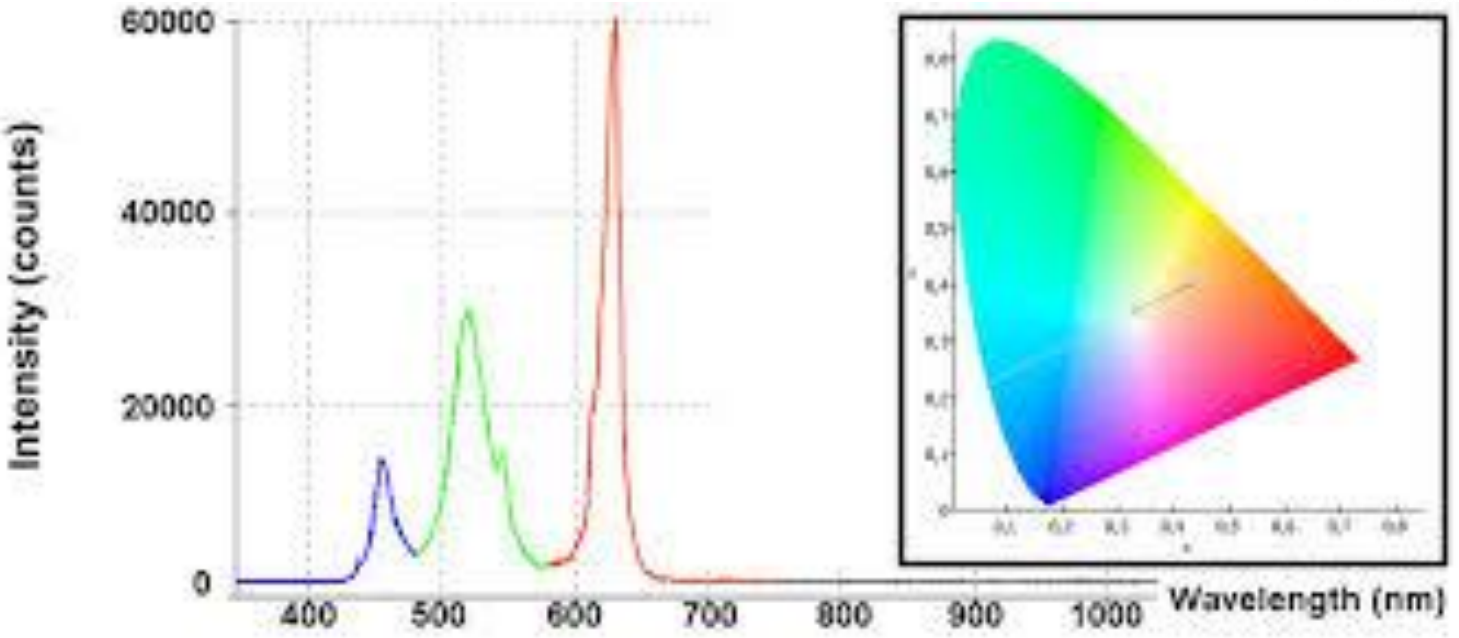


SPECTRUM OF COLORS

- Light appears to be white but …
- … it is made of many colors (!) …
- …usually coded into 7 colors, but many frequencies are present.



Cameras usually acquire 3 colors: R,G,B

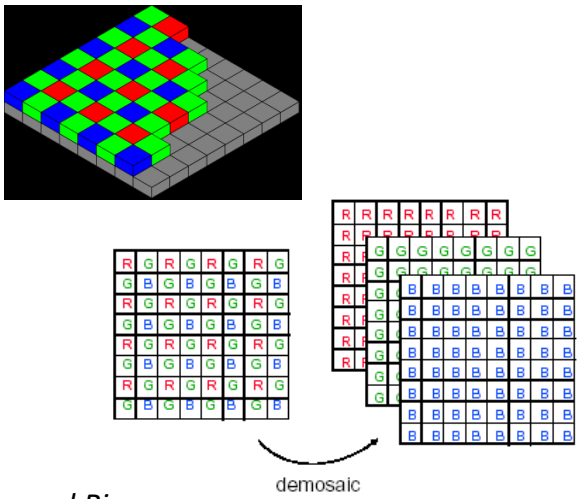


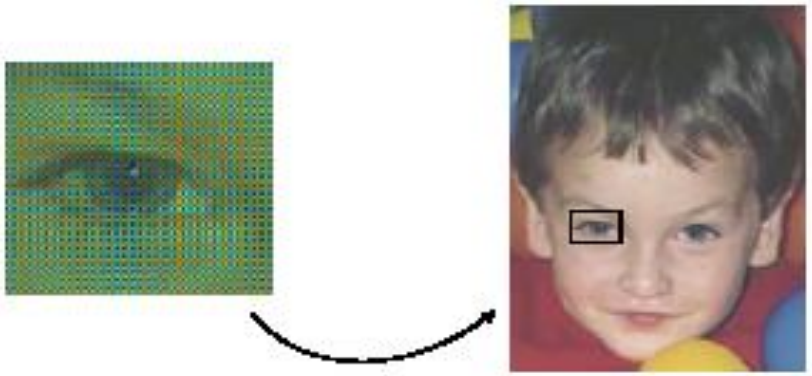
COLOR SENSORS

- Sensors can not distinguish colors (i.e., wavelength/photons' energy)
  - Count photons' number
  - CCD/CMOS matrixces generates images in grayscale. To get colors, we need filters.
- Two possible technologies:

1. **3-chip color**: incoming light is divided intpo the three components (R,G,B) using filters and prisms. Three different sensors are then used.

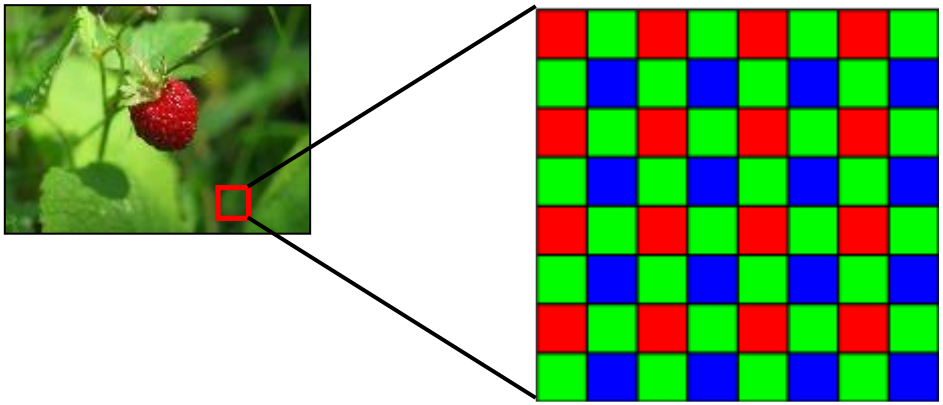
2. **Single-chip color**: uses color filters before the acquisition, which separates the different components for each sensor. The complete information is then reconstructed by firmware.



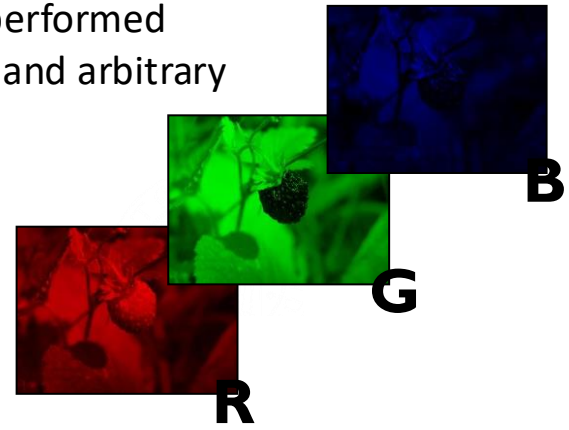


CFA INTERPOLATION DETECTION

Digital cameras acquire a single frequency component per pixels according to a regular pattern: Bayer mask



Every vendor use its own bayer masks, and the interpolation is performed with proprietary and arbitrary hardware



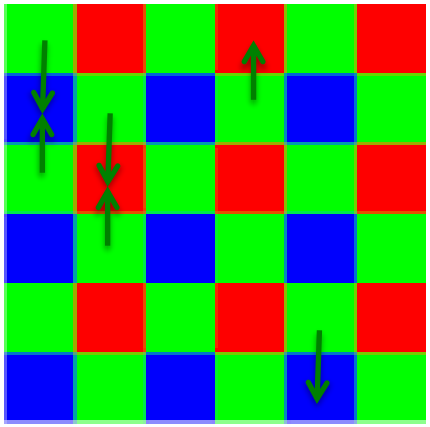
10		12		14	
	9		12		10
12		18		19	
	4		6		10
6		10		13	
	7		8		11



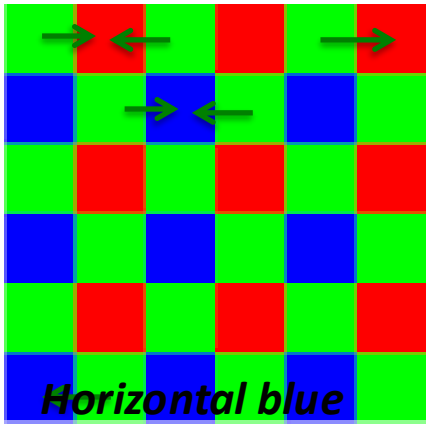
10		12		14	
11	9		12		10
12		18		19	
9	4		6		10
6		10		13	
	7		8		11

COLOR INTERPOLATION

Vertical green

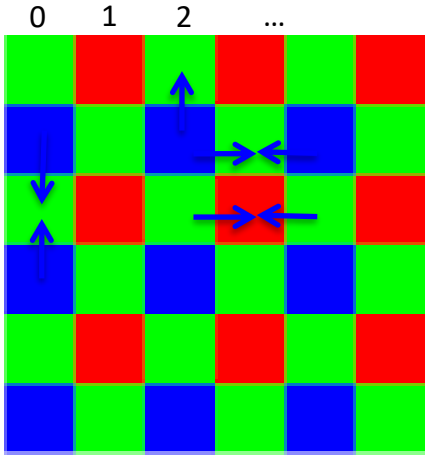


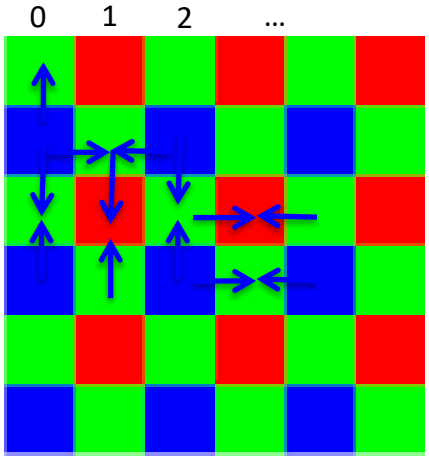
Horizontal green



Horizontal blue

Vertical blue





DEMOSAICING ARTIFACTS



Original



bilinear



Method A



Method B

SIMILARLY…

- We can distinguish images taken by a camera and by a scanner
- We can understand whether an image or a portion of an image has been resized.



Interpolation can be inferred from traces left on the image

11	3	15	12	14	14
11	9	13	12	16	10
12	8	18	21	19	18
4	4	7	6	12	10
7	5	13	6	45	9
11	7	10	8	13	11

6x6 pixel

11	7	3		15		12		14		14	
11	10	9		13		12		16		10	
12	10	8		18		21		19		18	
4	4	4		7		6		12		10	
7	6	5		13		6		45		9	
11	6	7		10		8		13		11	

6x12 pixel

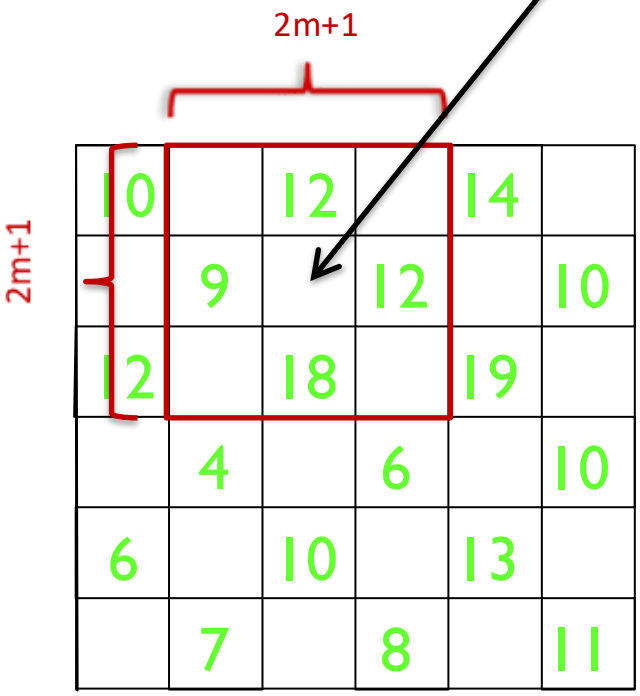
A SIMPLE ESTIMATION APPROACH

- simple linear model for the periodic correlations introduced by CFA interpolation: weighted sum of pixels in a small neighborhood
- both easy to parametrize and can reasonably approximate CFA interpolation algos
- Let's treat colors independently
- If correlations is known (i.e., the parameters of the linear model), then it would be straightforward
- There are unknown information: whether pixel is interpolated or not!

DEMOSAICING DETECTION

Let's consider the interpolated signal $\hat{I}_d(x,y) = \sum_{u=-m}^{m} \sum_{v=-m}^{m} \alpha_{u,v} I(x+u, y+v)$

And the residual $r(x,y) = I(x,y) - \sum_{u=-m}^{m} \sum_{v=-m}^{m} \alpha_{u,v} I(x+u, y+v)$



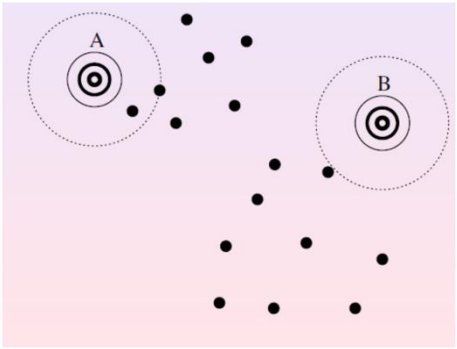
The parameters of the interpolation $\alpha_{u,v}$ define the model/filter that performs the interpolation.

Finding parameters $\alpha_{u,v}$ ≡ identifying the interpolator/camera model

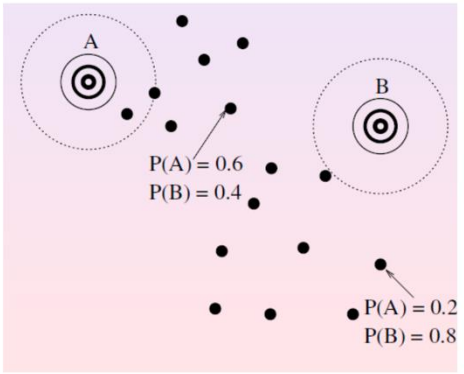
Define this as probability of observing r from the model instead from another model (outlier)

EXPECTATION-MAXIMIZATION: MAIN DESCRIPTION

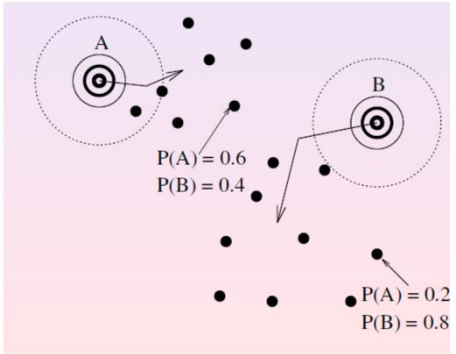
Suppose you have a set of points that can be generated by two sources A or B (Gaussians)



- E-step: estimate the probability for each point



- M-step: recompute the models



EXPECTATION-MAXIMIZATION

- Method to find model with missing data
- We do not know how to distinguish std. pixels or interpolated with the unknown method
- EM solve: introduces unknown or hidden variables

Maximize $\mathrm{p}[x;\theta] = \sum_z p[x;z,\theta]$ leading to log-likelihood for samples x_i

$$\ell(\theta) = \sum_i \log p[x_i;\theta] = \sum_i \log \sum_{z_i} p[x_i;z_i,\theta]$$

From Bayes we have

$$log\ p[x;\theta] = log \sum_z p[x;z,\theta] = \log \sum_z q(z) \frac{p[x;z,\theta]}{q(z)} \geq \sum_z q(z) \log \frac{p[x;z,\theta]}{q(z)}$$

That leads to

$$log\ p[x;\theta] \geq \sum_z \sum_i q(z_i) \log \frac{p[x_i;z_i,\theta]}{q(z_i)}$$

$q(z)$ is the posterior distribution; typically $q(z) \sim p[z_i|x_i,\theta]$

EXPECTATION-MAXIMIZATION

$$log\ p[x;\theta] \geq \sum_z \sum_i q(z_i) \log \frac{p[x_i; z_i, \theta]}{q(z_i)}$$

$q(z)$ can be approximated as

$$q(z) \sim p[z\ |x,\theta] = \frac{p[x,z|\theta]}{p[x]} = \frac{p[x,z|\theta]}{\sum_z p[x\ |z]\ p[z]} = \frac{p[x\ |\ z,\theta]\ p[z]}{\sum_z p[x\ |z]\ p[z]} = \frac{p[x\ |\ mod]\ p[mod]}{\sum_z p[x\ |mod]\ \ p[mod] + p[x|out]\ p[out]}$$

Being model and outlier equally probable

$q(z) \sim \frac{p[x\ |\ mod]}{\sum_z p[x|mod] + p[x|out]]} = \frac{L(mod)}{L(mod)+L(out)}$ distributes values between model/outlier

Then, algorihm becomes

1) E-step: compute $\sum_i p[z_i\ |x_i, \theta_i] = \frac{L(mod)}{L(mod)+L(out)}$

2) M-step: maximize $\sum_z \sum_i q(z_i) \log \frac{p[x_i;z_i,\theta]}{q(z_i)}$

Repeat multiple times until convergence

TUTORIAL ON
https://www.youtube.com/watch?v=3zbAsgCf1Sw

DEMOSAICING DETECTION

Assuing that the probability of observing a sample generated by weights $\alpha = [\alpha_{u,v}]$ is Gaussian and the probability of other is uniform $1/\delta$, the prior is

$$w(x,y) = P[\alpha | r(x,y)] = \frac{e^{-r^2/\sigma}}{e^{-r^2/\sigma} + 1/\delta}$$

where $1/\delta$ is uniform probability of observing a different model or authentic pixels (see the previous E-step)

It is possible to compute this with an EM estimation

E-step

$$r(x,y) = I(x,y) - \sum_{u=-m}^{m} \sum_{v=-m}^{m} \alpha_{u,v} I(x+u, y+v)$$

M-step: compute weights and minimize

$$E(\boldsymbol{\alpha}) = \sum_{x=m+1}^{n-m} \sum_{y=m+1}^{n-m} w(x,y) \left(I(x,y) - \sum_{u=-m}^{m} \sum_{v=-m}^{m} \alpha_{u,v} I(x+u, y+v) \right)^2$$

EXAMPLES

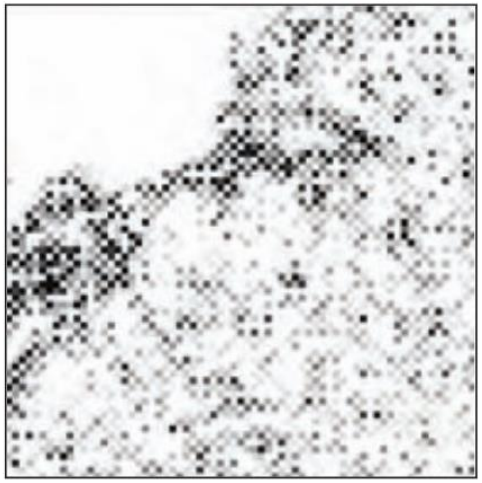
CFA interpolated

(b) probability

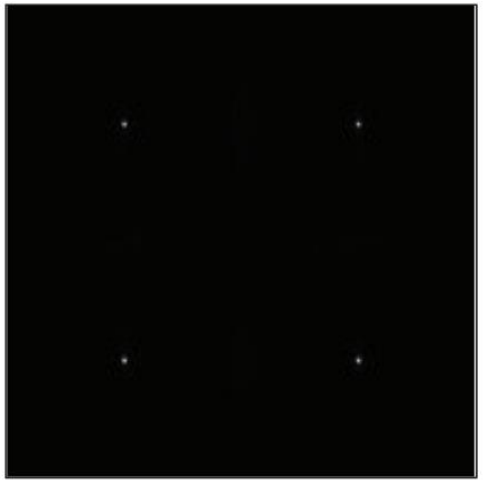
(c) Fourier of probability



(a)



(b)

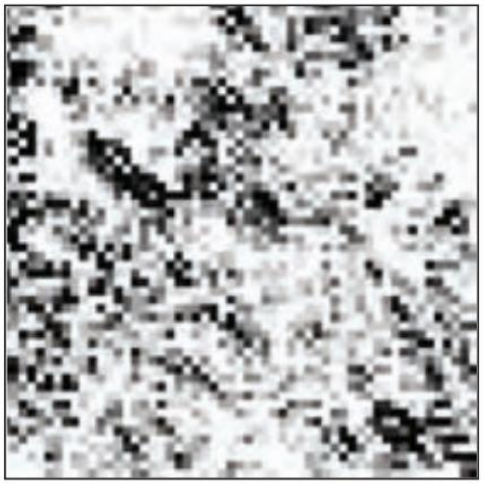


(c)

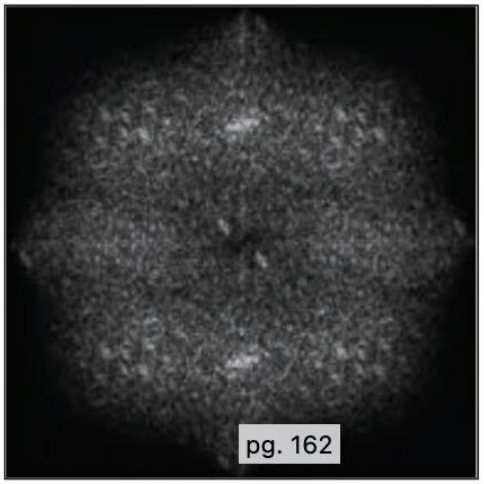
CFA not interpolated



(a)



(b)



(c)

COMPRESSION

CODING

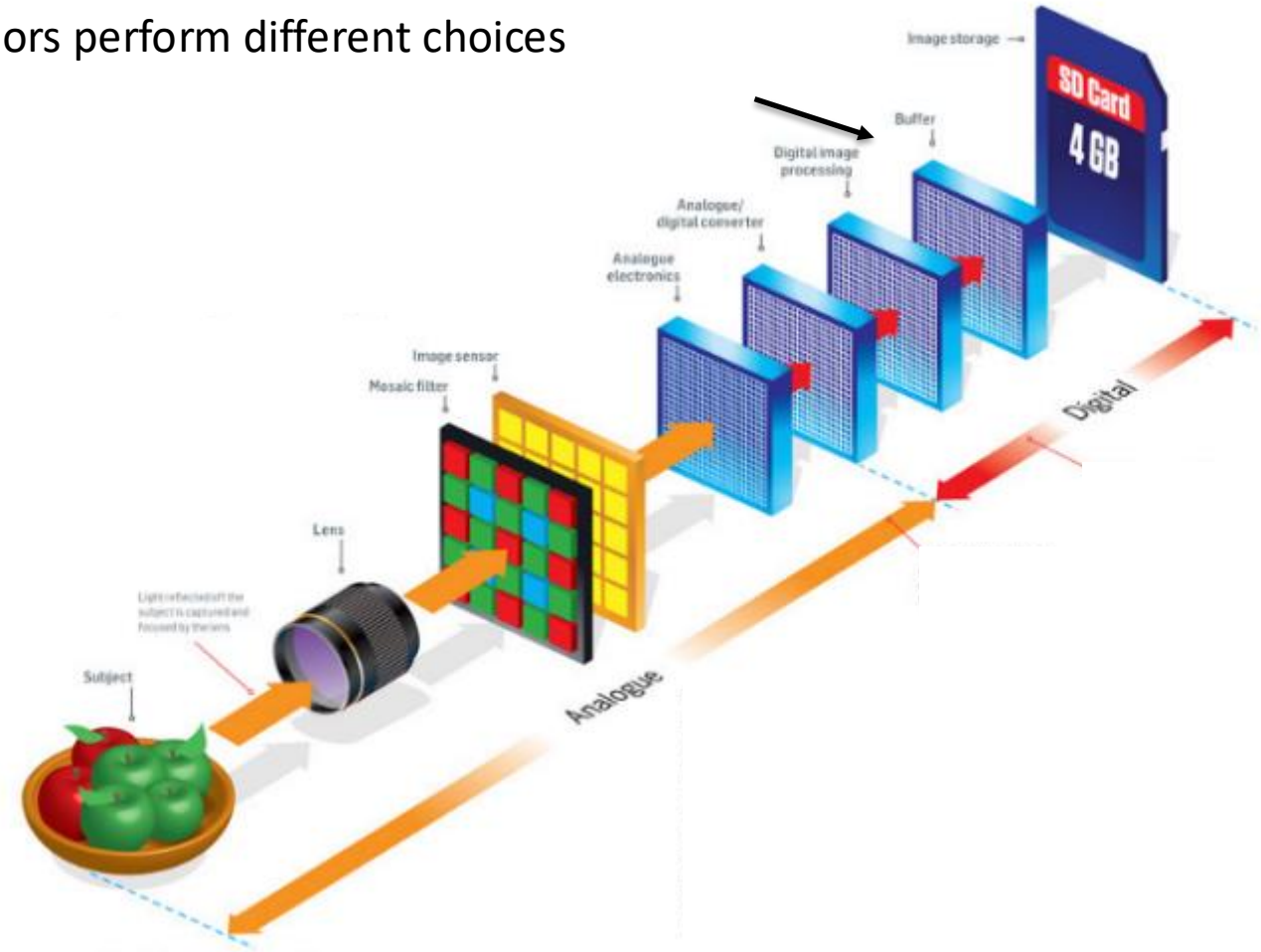
Images and videos are usually compressed.

Coding standards define bit stream structure and decoding operations

Encoder is free; different vendors perform different choices

Codec related features!

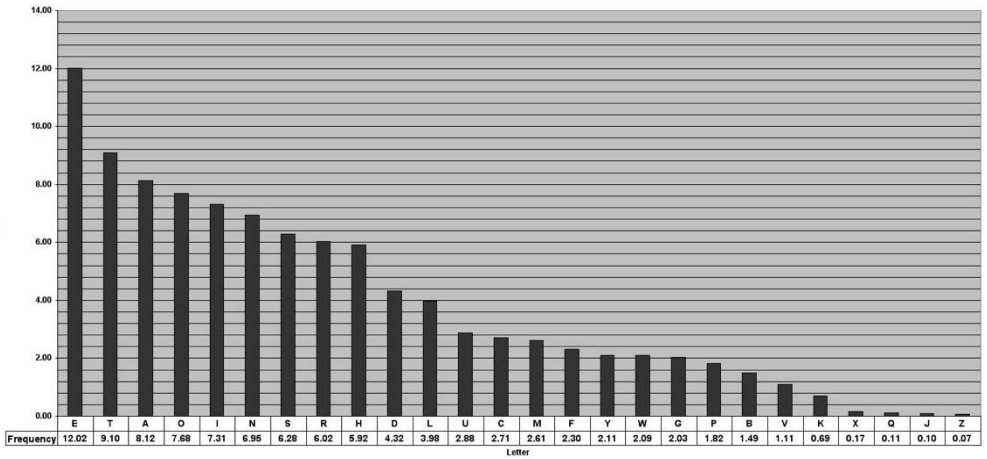
(Image, Video, Audio)



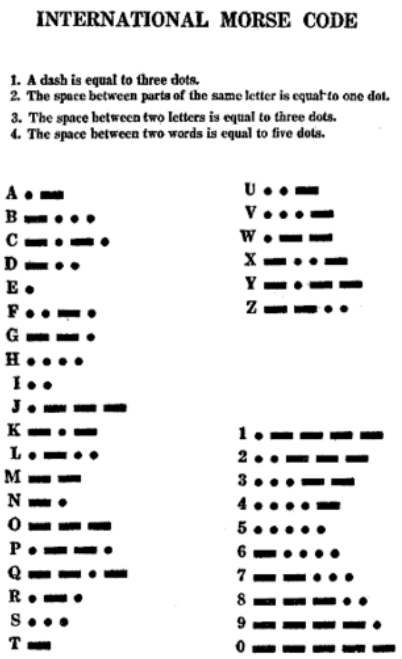
LOSSLESS AND LOSSY CODING

Lossless

- Data are converted into a binary bit stream;
- mapped into non-ambiguous strings
- Sources are perfectly reconstructed, no distortion is present.



	E	T	A	O	I	N	S	R	H	D	L	U	C	M	F	Y	W	G	P	B	V	K	X	Q	J	Z
Frequency	12.02	9.10	8.12	7.68	7.31	6.95	6.28	6.02	5.92	4.32	3.98	2.88	2.71	2.61	2.30	2.11	2.09	2.03	1.82	1.49	1.11	0.69	0.17	0.11	0.10	0.07



Lossy

- Obtaining a high-fidelity reconstruction (as much as possible) given a maximum limit to the maximum amount of coded data (bit rate);
- Given a target distortion level between the original source and the reconstructed one, generate a bit rate size as small as possible.
  - Not invertible
  - High compression ratio

LOSSY CODING

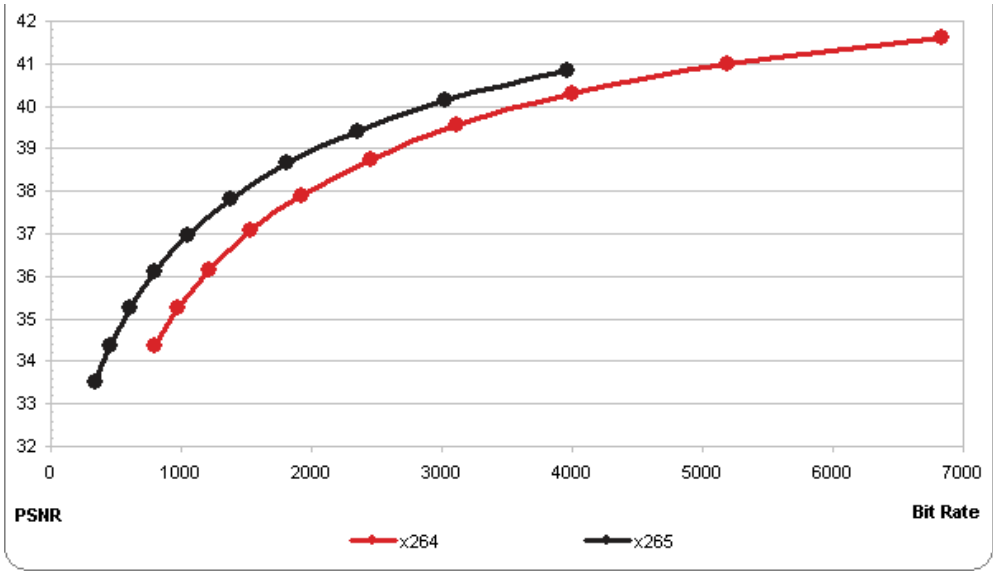
Objective measures

$$e = I(x,y) - I_R(x,y)$$

$$PSNR = 10\ log_{10}\frac{255^2}{E[e^2]}$$

$$SSIM(x,y) = \frac{(2\ \mu_x\ \mu_y + c_1)\ (2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)\ (\sigma_x^2 + \sigma_y^2 + c_2))}$$

Subjective measures
(perceptual quality tests)



Lossy coding is interesting since compression leaves misprints on the image/audio/video that can be used to trace the origin of the content.

Lossy image format
- JPEG
- Gif
- JP2000

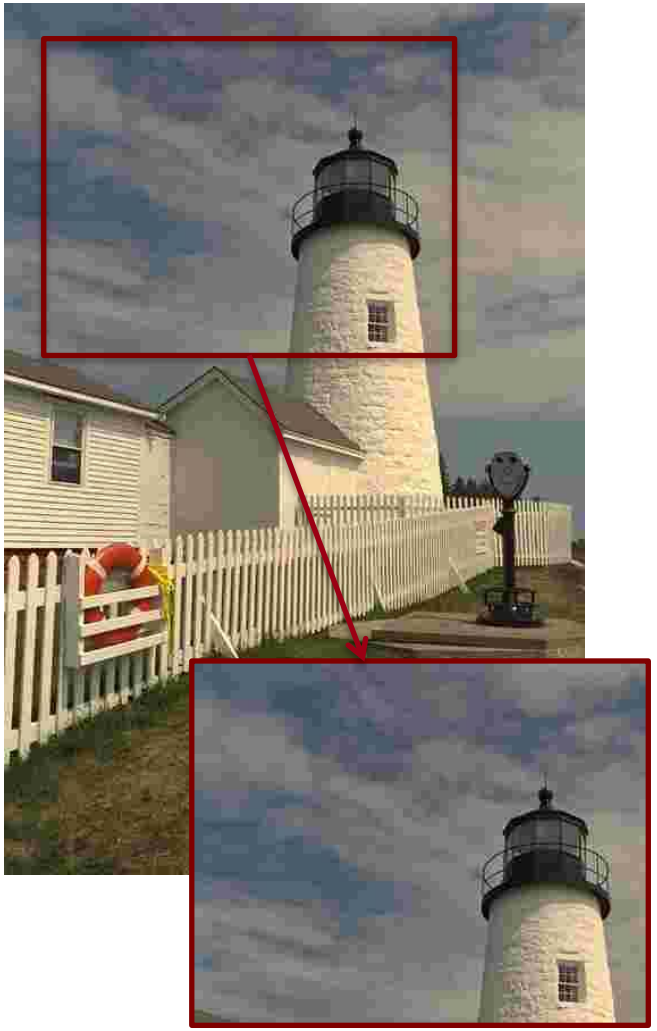
Lossy video format
- MPEG 1,2,4
- H.26x x=4,5 or AVC,HEVC
- Flv,Ogg,Webm

EXAMPLE

QF=90 PSNR=41.26 bpp=2.31
Compression ratio=1:10

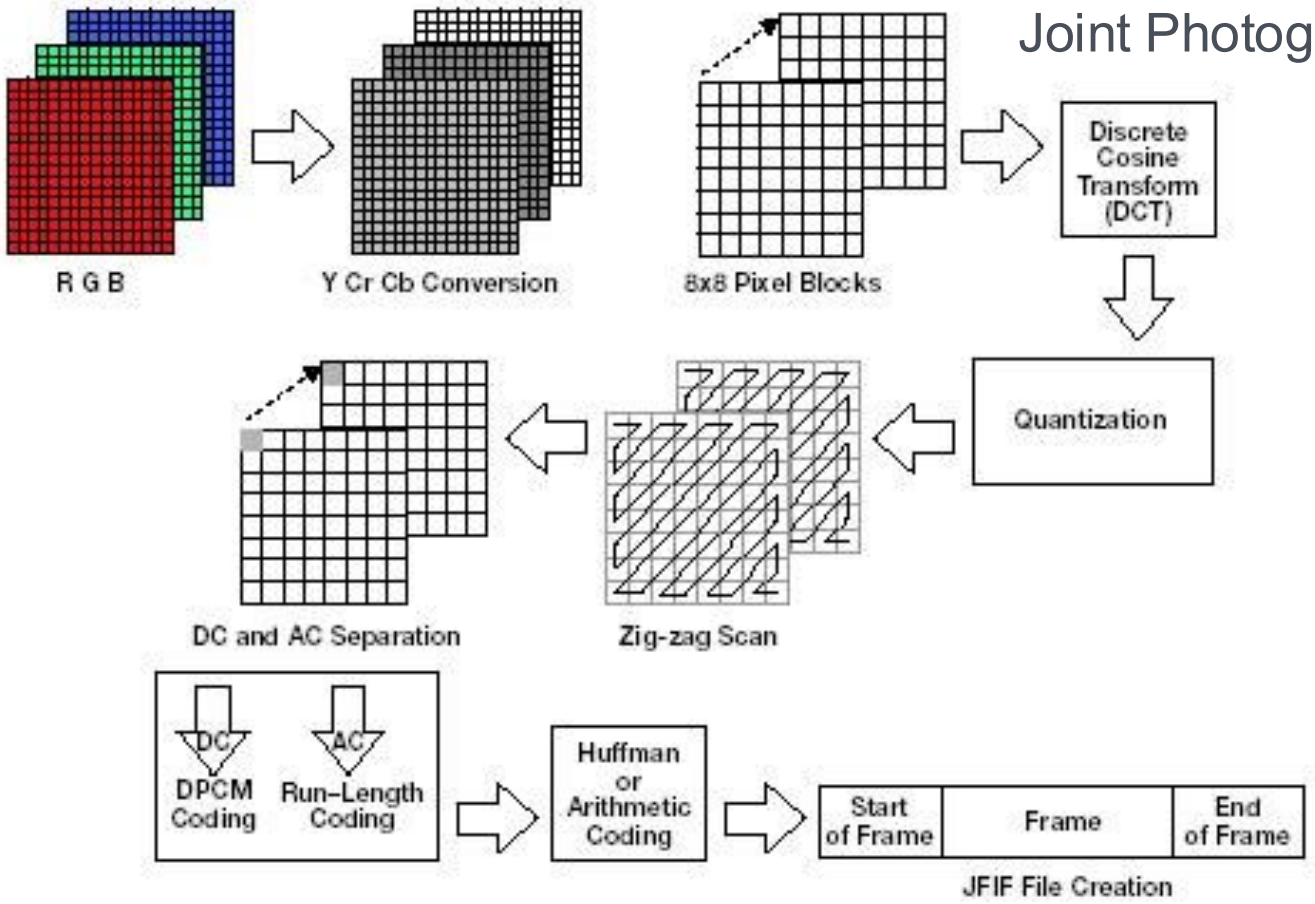


QF=10 PSNR=31.98 bpp=0.31
Compression ratio=1:77



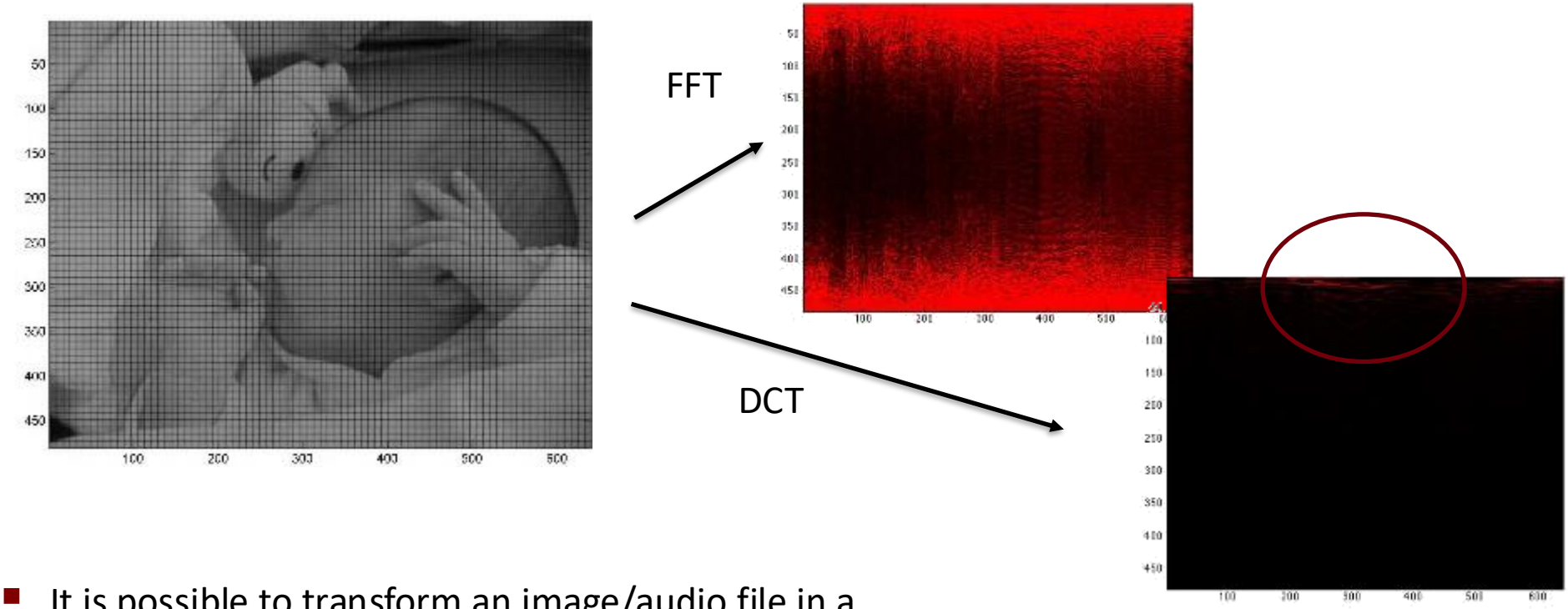
JPEG CODING SCHEME

Joint Photographic Experts Group

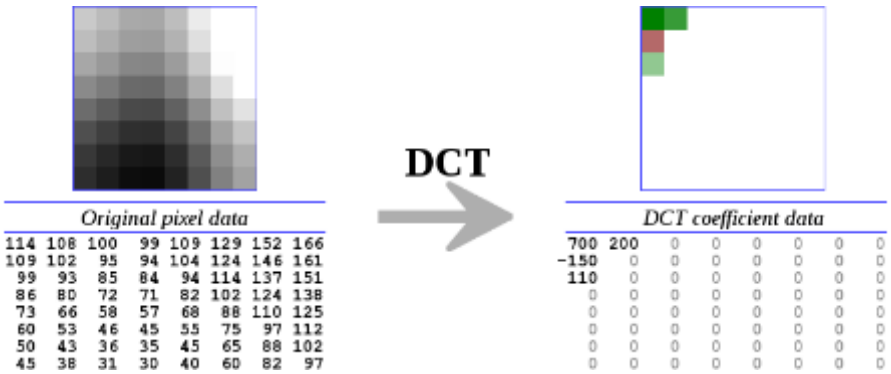


1. RGB-YUV Conversion + subsampling
2. Block partitioning
3. DCT
4. Quantization
5. Entropy

FREQUENCY REPRESENTATION

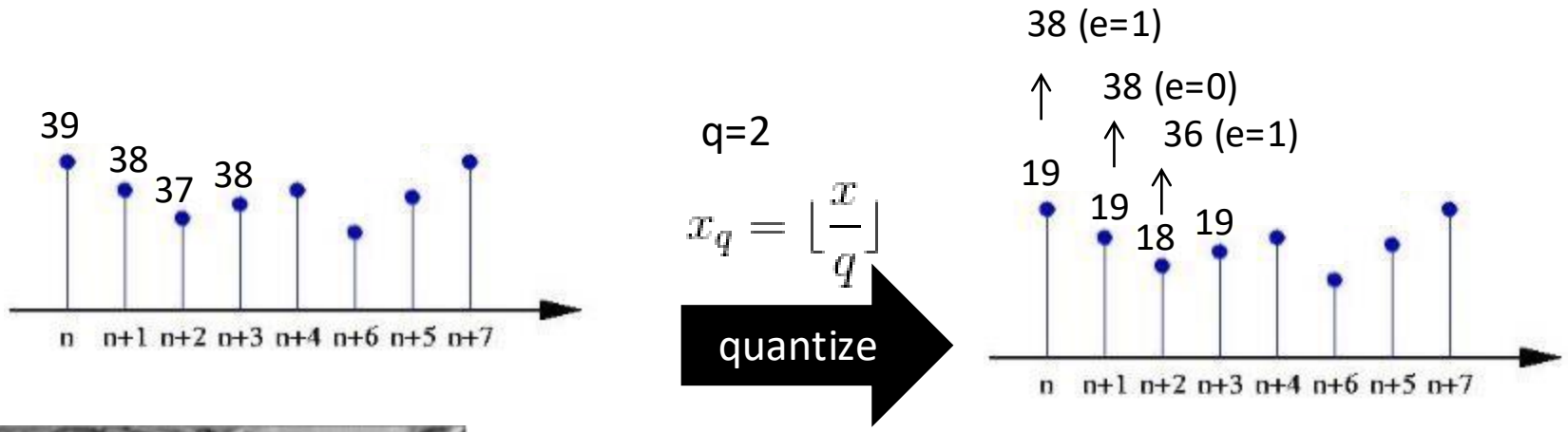


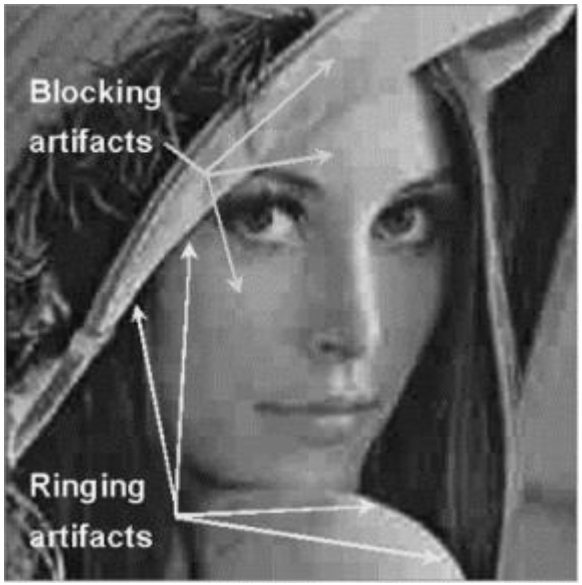
- It is possible to transform an image/audio file in a representation that is more suitable for the compression
- Frequency-based representation
- Usually operated on blocks



QUANTIZATION & BLOCKING ARTIFACTS

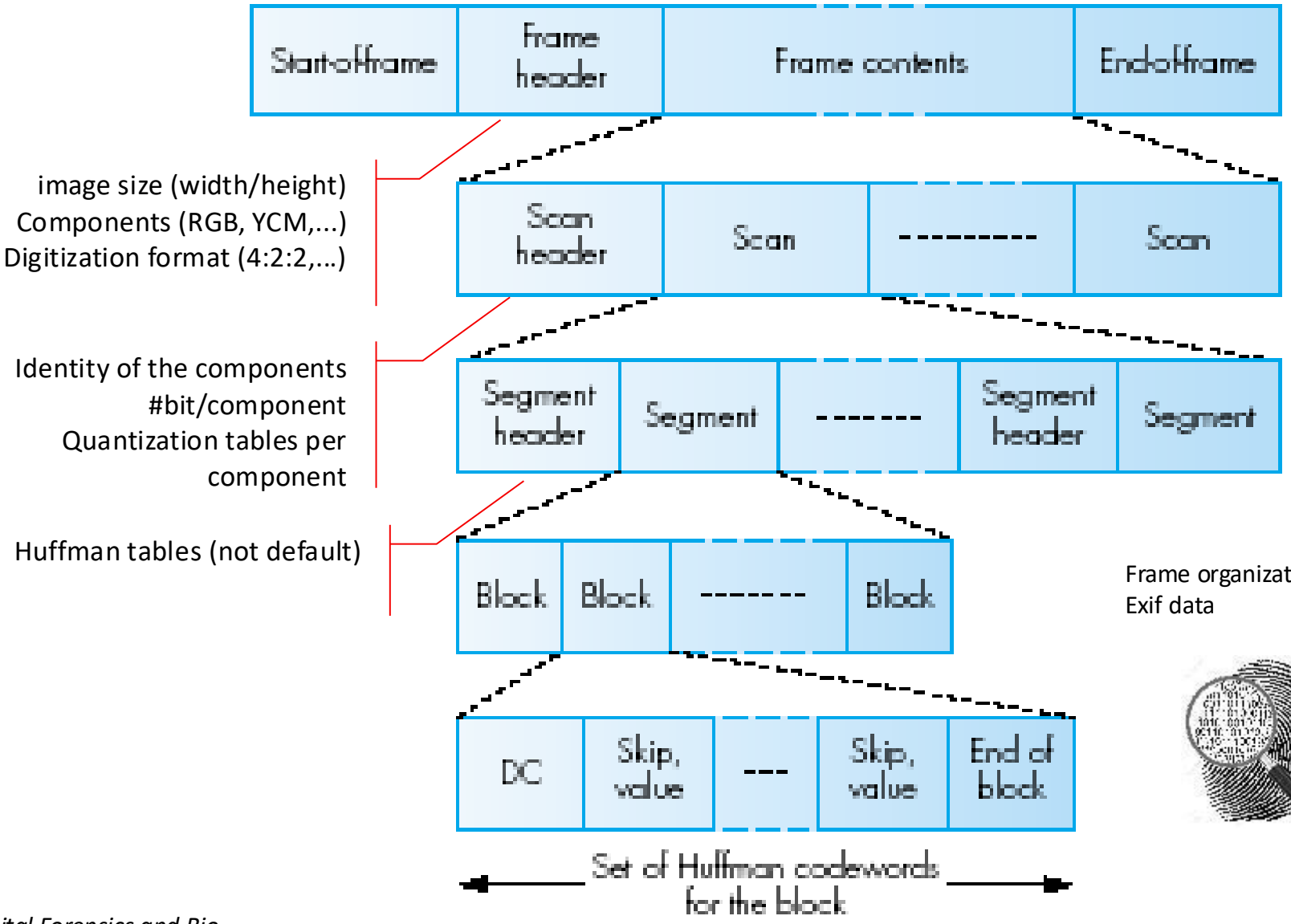
Quantization introduce information loss. But permits reducing the bit rate!





- The quantization of decorrelated data introduces image distortion, called *blocking artifacts*
- Since coders operate on square blocks, quantization artefacts have a square shape.
- Many coefficients quantized to zero – *ringing artifacts*
- The more aggressive the quantization, the more evident the blocking effect!
- Bit rate limits impose stronger artifacts if the video/image is difficult to code.

FRAME BUILDING

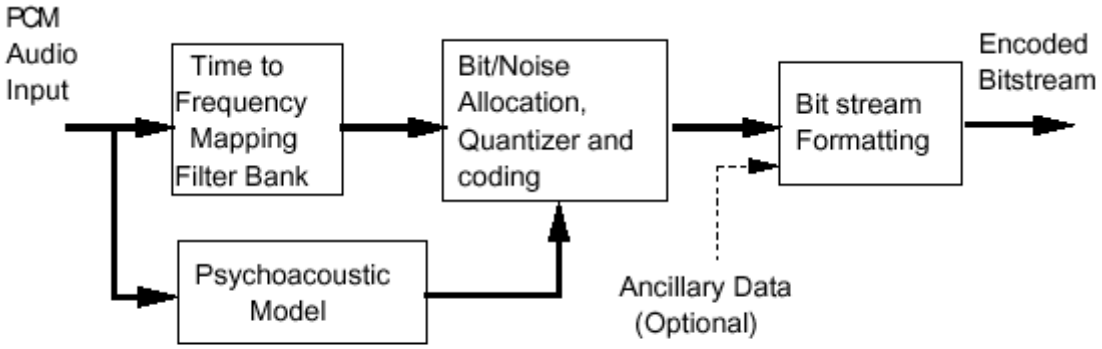


Frame organization and Exif data



SIMILAR STRATEGIES FOR AUDIO: MP3

- MPEG-1 Audio Layer 3 (MPEG = Moving Picture Experts Group)
- Early '90s
- Music, audio tracks, recordings
- Popular on the Internet

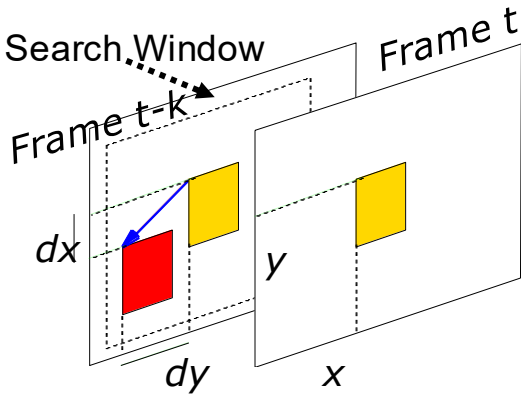


MPEG/Audio Encoder



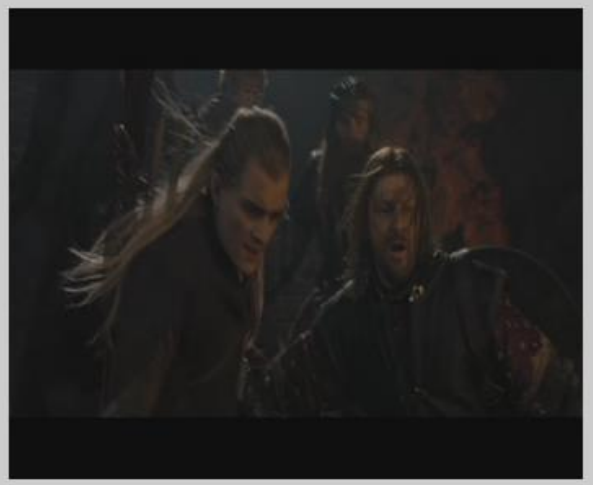
Frame division, frequency optimization and compression, pseudo-acoustic models

GENERIC PREDICTIVE VIDEO CODING ARCHITECTURES

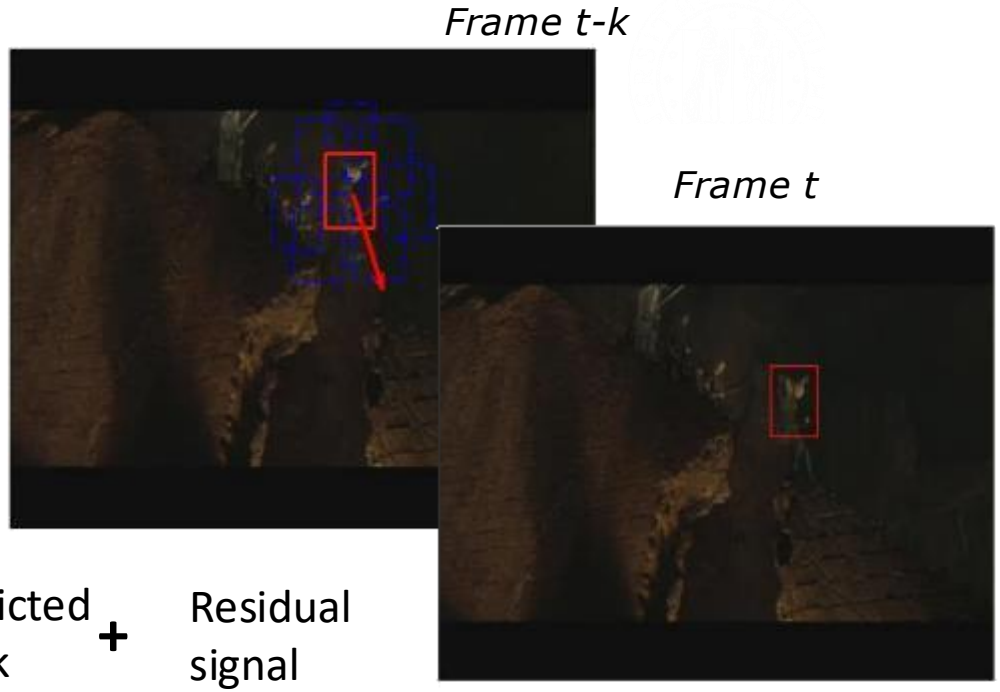


Adjacent frames in a video sequence are correlated: there is a lot of redundant information

It is possible to reduce the redundancy predicting the information: only the difference needs to be coded.



Current block = Predicted block + Residual signal



EXAMPLE: MC-PREDICTION FOR TWO CONSECUTIVE FRAMES



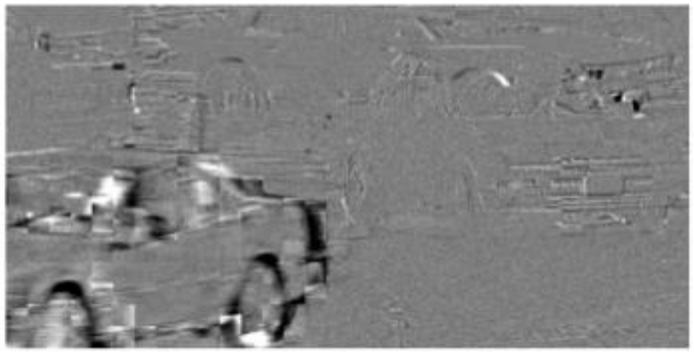
Reference Frame



Current Frame (to be compressed)



Prediction of current frame



Prediction error

CONSTANT BIT RATE

CBR 250kbit/s

Rate control/optimization algorithms are among the digital fingerprints





CBR 1Mbit/s





CONSTANT QUALITY

CRF=45 size=665KB



size=1.2MB



CRF=27 size=2.0MB



size=5.3MB

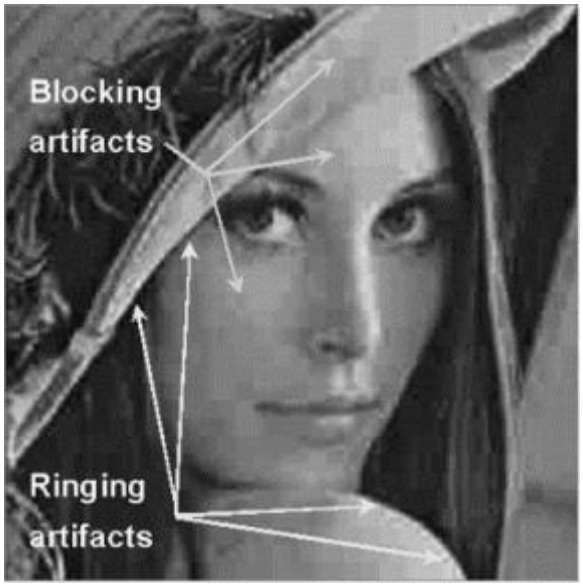


CODING CURRENT IMAGE AND VIDEO COMPRESSION STANDARDS

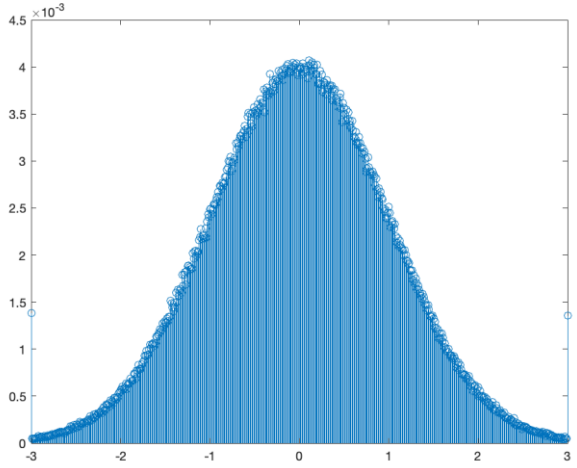
Standard	Application	Bit Rate
JPEG	Continuous-tone still-image compression	Variable
H.261	Video telephony and teleconferencing over ISDN	p x 64 kb/s
MPEG-1	Video on digital storage media (CD-ROM)	1.5 Mb/s
MPEG-2	Digital Television	2-20 Mb/s
H.263	Video telephony over PSTN	33.6-? kb/s
MPEG-4	Object-based coding, synthetic content, interactivity	Variable
JPEG-2000	Improved still image compression	Variable
H.264 / MPEG-4 AVC	Improved video compression	10's to 100's kb/s

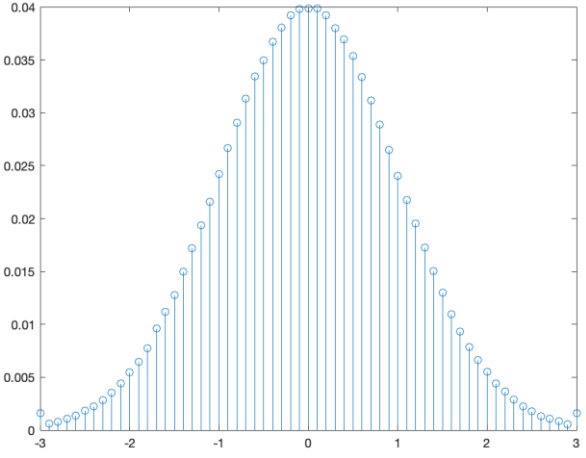
JPEG COMPRESSION FOOTPRINTS

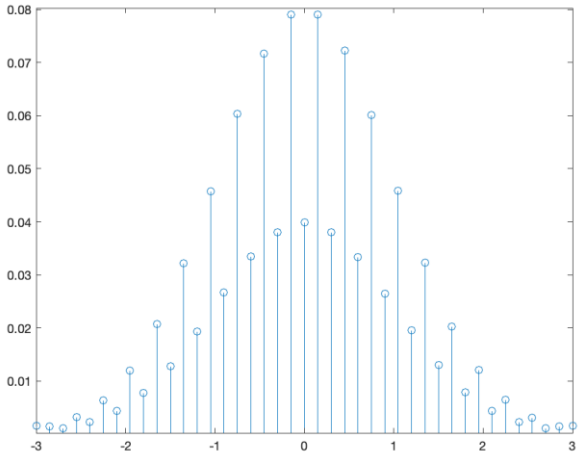
- Detection of JPEG coding footprints
  - Blocking artifacts
    - [Liu and Heynderickx, ICASSP 2008]
  - DCT coefficient statistics
    - [Fan and de Queiroz, TIP 2003]



- Image codec detection (JPEG / wavelet / DPCM)
  - [Lin et al., TIFS 2009]





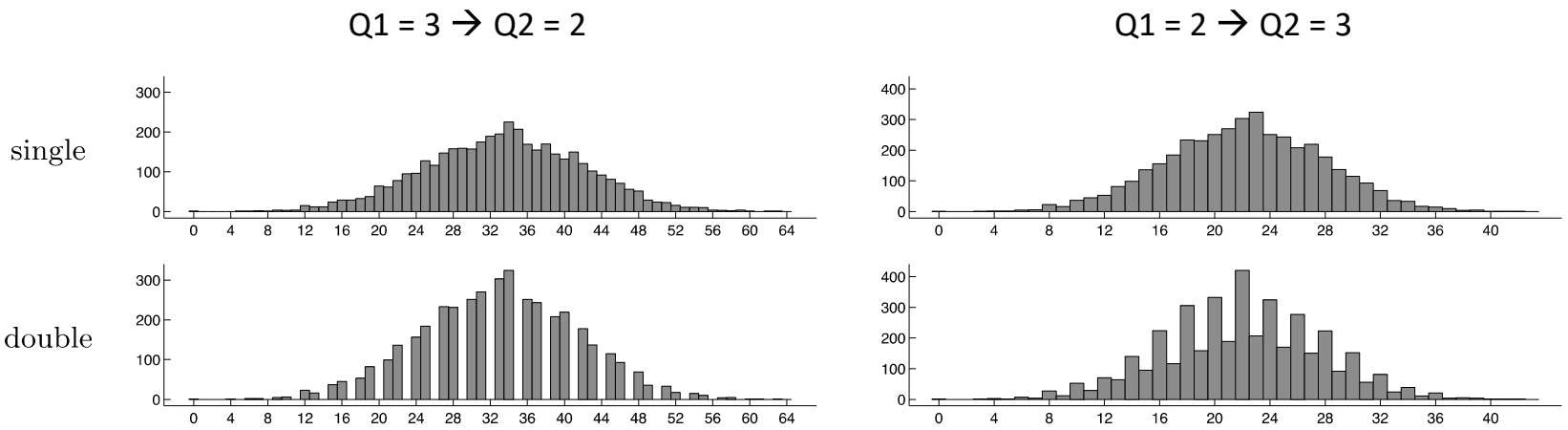


DOUBLE (MULTIPLE) JPEG COMPRESSION

The goal is to identify traces of previous compression and, possibly, to estimate the codecs

Detect if single or double compression was applied (and which were the parameters)

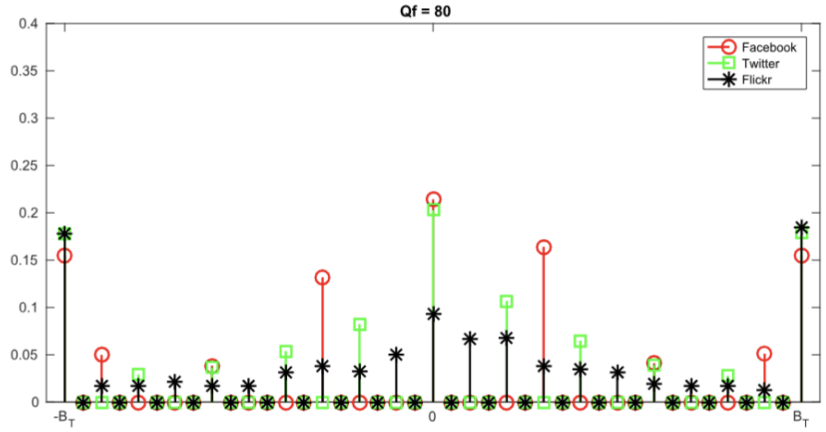
- DCT coefficient statistics
  - [Lukas and Fridrich, DFRWS2003][Bianchi and Piva, WIFS 2011]
- First digit's statistics
  - [Fu et al., SPIE 2009][Milani et al., ICASSP 2012]

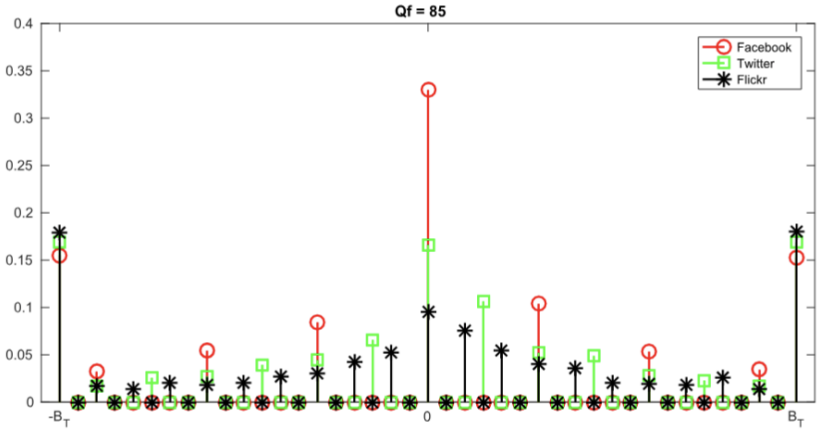


[Wang and Farid, ACM 2006]

SOCIAL MEDIA STATISTICS

- Images, during upload on SN, are usually re-compressed.
- Different quantization strategies adopted -> statistics change



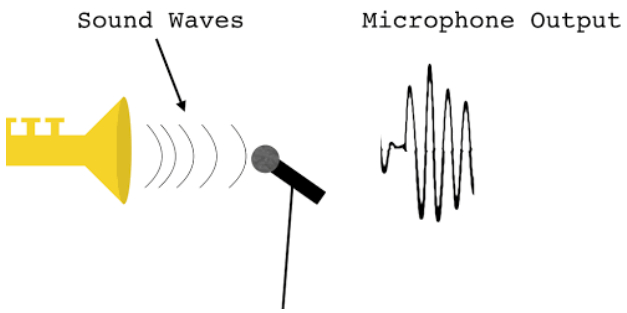


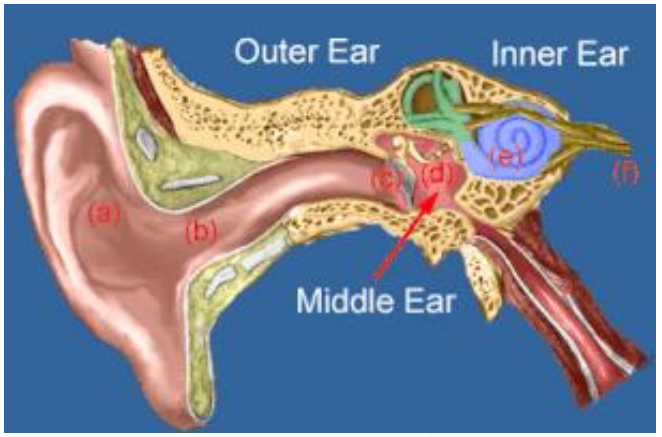
- Neural network can distinguish image provenance (FusionNet, 2019)

Classification (%) vs SNs	Facebook	Flickr	Google+	Instagram	Original	Telegram	Twitter	WhatsApp
Facebook	**91.31**	6.21	0.00	0.08	2.40	0.00	0.00	0.00
Flickr	0.90	**86.77**	0.03	0.18	3.26	0.70	8.14	0.02
Google+	0.01	0.03	**88.01**	0.48	11.44	0.02	0.00	0.02
Instagram	0.40	0.00	0.00	**98.80**	0.80	0.00	0.00	0.00
Original	0.00	0.00	0.00	0.00	**99.01**	0.99	0.00	0.00
Telegram	0.01	0.00	0.00	0.00	1.12	**98.87**	0.00	0.00
Twitter	0.11	2.00	0.00	0.00	1.51	0.11	**96.27**	0.00
WhatsApp	0.00	0.12	0.00	0.03	0.72	0.00	0.00	**99.13**

DIGITAL AUDIO RECORDERS

DIGITAL AUDIO

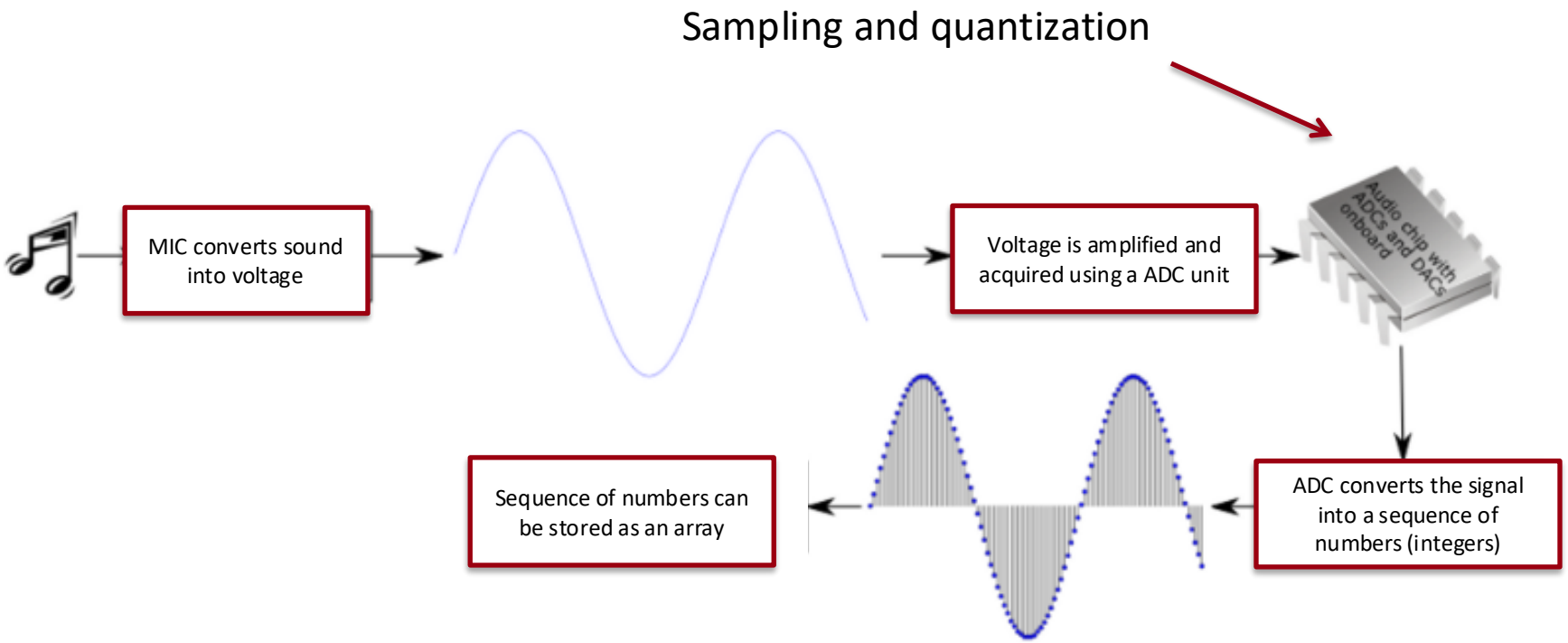




- **Sound** is a vibration that propagates as an acoustic wave, through a transmission medium such as a gas, liquid or solid.
- This vibration can be modelled as a wave signal that propagates through space
- Whenever the wave encounters an acoustic receiver (microphone, human ear, …) an acoustic membrane start oscillating.
- In a microphone, oscillation is converted into a voltage
- In a human ear, vibration is converted into an electric stimulus
- Voltage signal can be sent to a loudspeaker → reverse process
- Voltage is converted into a vibration of speaker's cone → vibration back in the air

DIGITAL AUDIO ACQUISITION

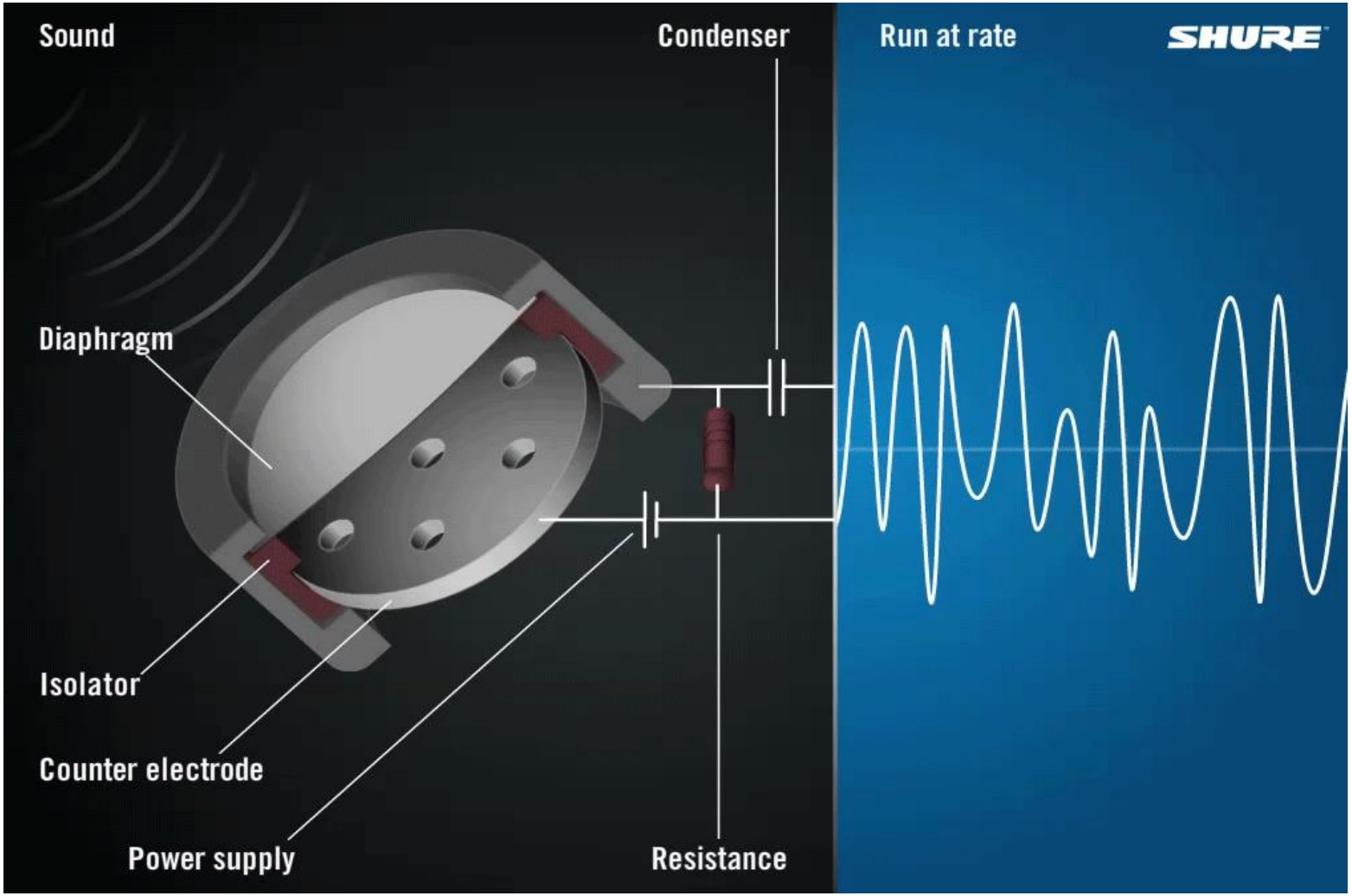
After acquisition, audio (voltage) waves need to be stored in some numerical format



Samples are controlled by the acquisition (sampling) frequency F_s

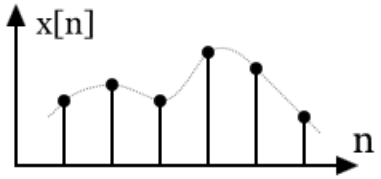
e.g. CD quality F_s=44100 Hz = 44.1 KHz phone communications 8000 Hz = 8 KHz

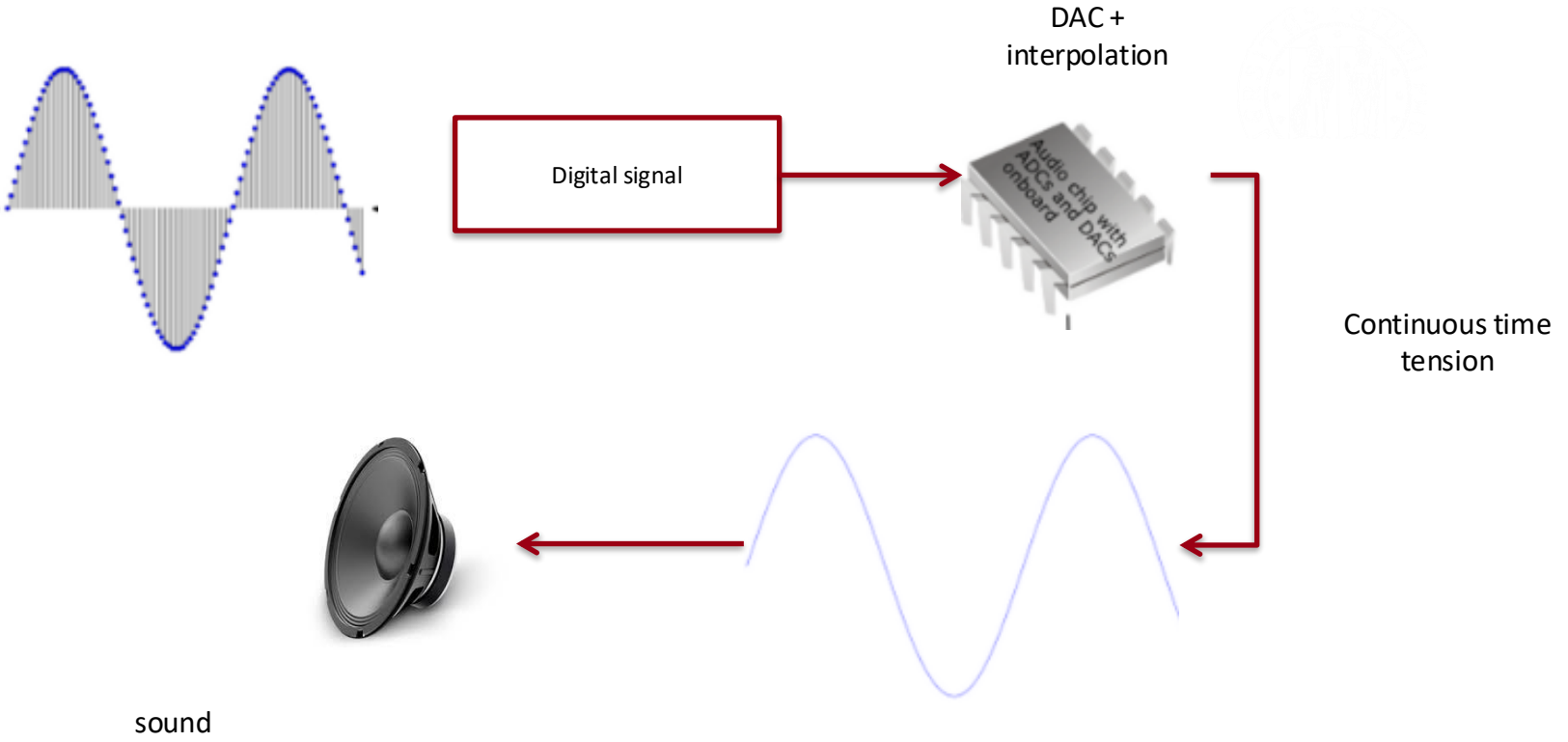
FROM SOUNDWAVE TO VOLTAGE



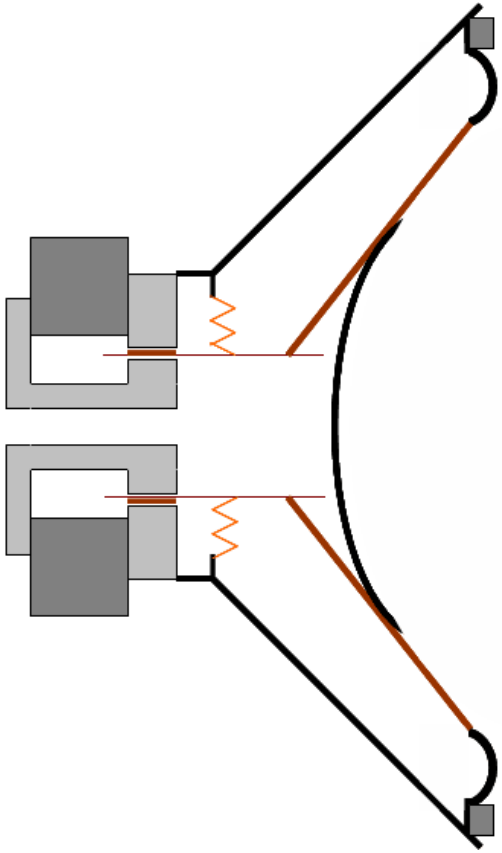
DIGITAL AUDIO REPRODUCTION

PC sound card performs the opposite operation



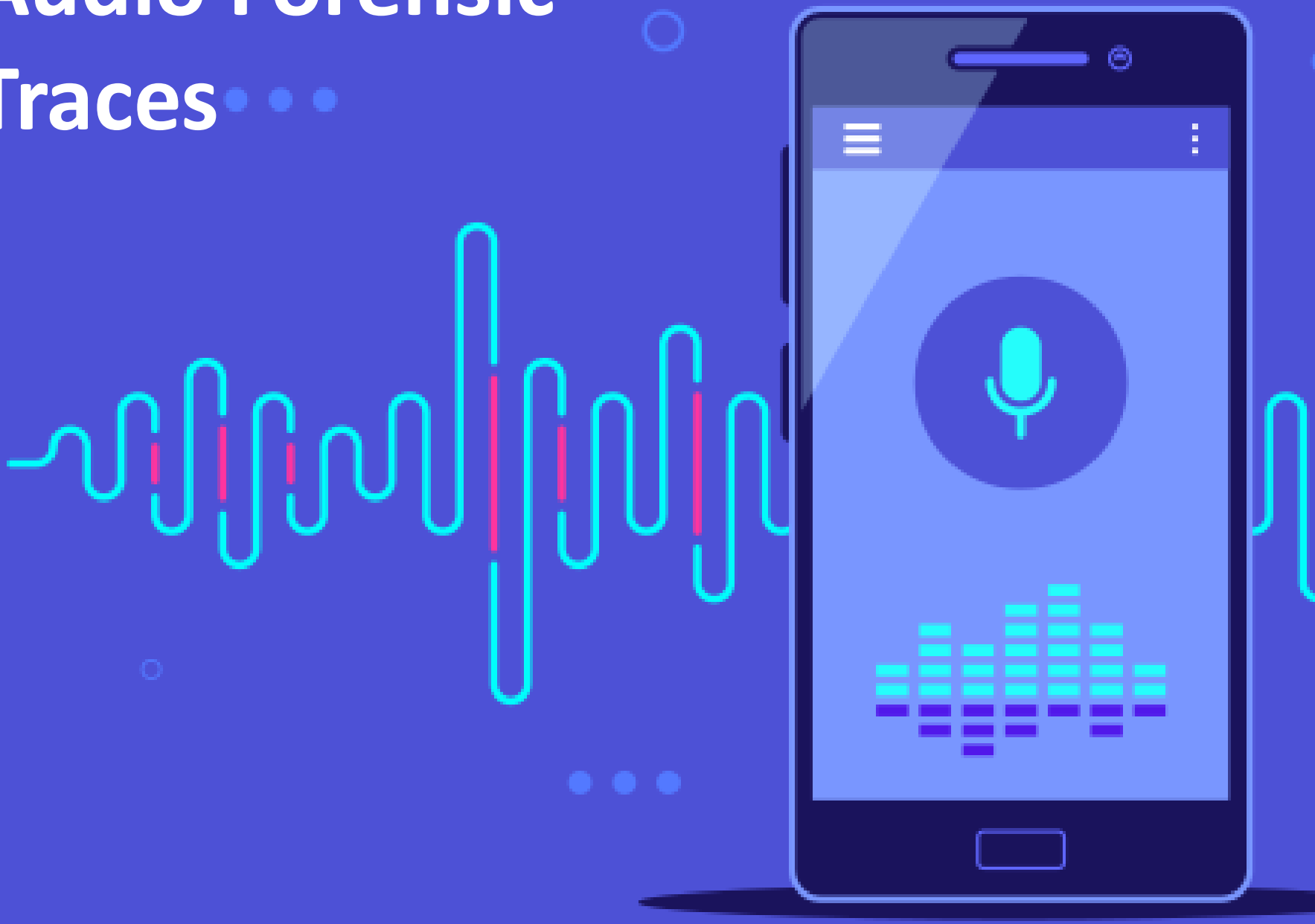


VICEVERSA





Audio Forensic Traces



AUTHENTICATION OF DIGITAL AUDIO

Used in criminal investigations such as bribery, political corruptions, drug deals, and other racketeering activities.

Evaluated for potential use as digital evidence in court

Integrity and **authenticity**

- Detection of discontinuities in the recorded signals as indicators for deletions
- Detection of specific attacks that violate the source characteristics, such as filter and compression, insertion or mixing, etc.

- Authentication of the microphone as the **acquisition sensor** (device)
- Authentication (entity authenticity) of other selected **contextual components**, such as the recording location (room characteristics), original or playback, background noises, etc

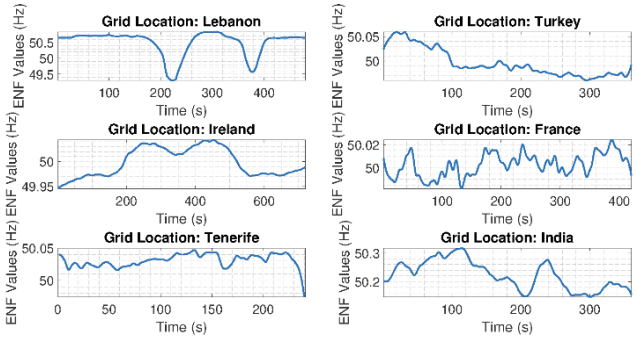
Analyzed features:

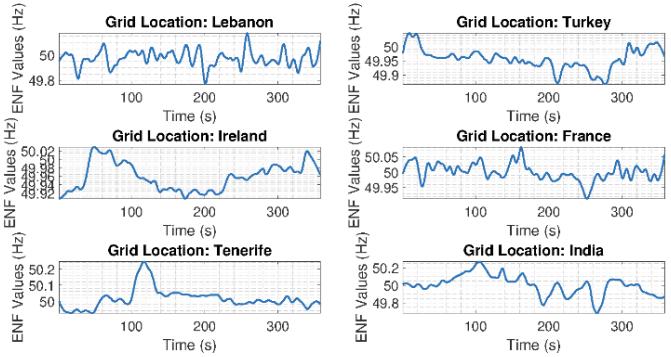
- Electric network frequency (ENF) of a recording setup,
- Content-based consistency analyses on local phenomena found in the time-domain representation of a recording
- Microphone response-based pattern recognition using acquisition setup nonlinearities

ELECTRICAL NETWORK FREQUENCY (ENF) ANALYSIS

In audio recordings, electrical power introduced a background hum; frequency changes in the hum are a fingerprint of a precise location.

Slight differences arealso present for different Rooms within a building





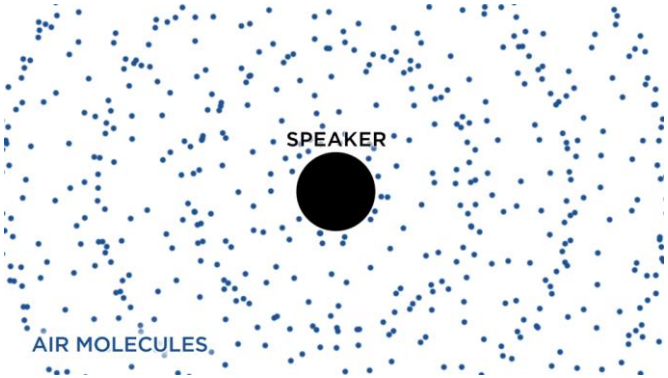
Sequence:

- Divide the sequence into overlapping frames
- Compute Fourier transform (STFT)
- Select spectrum for the different harmonics of the nominal frequency $f_0 = 50{,}60$ Hz. with an interval of f_B
- Created a weighted spectrum
  $S(f) = \sum_{k=1}^{L} w_k P_{B,k}(kf)$
- Compute features like mean values, variance, AR models ….
- Final classification using an ML classifier

A similar signal can be extracted from videos (artifical illumination)

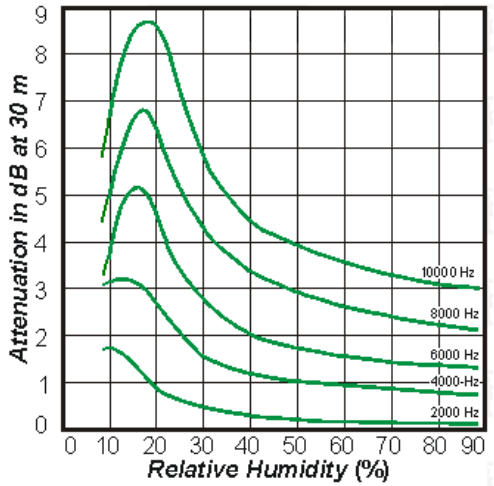
SOUND PROPAGATION

- Soundwave propagates through space at a certain speed (it takes some time to hear the sound)

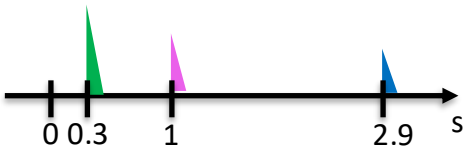


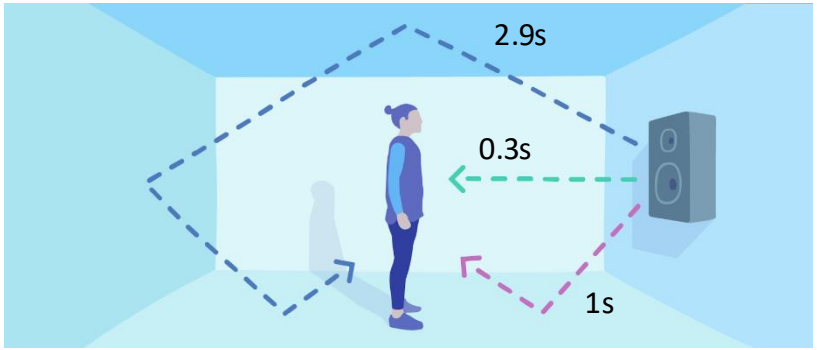
Substance	Temp (°C)	Speed (m/s)
Gases		
Carbon Dioxide	0	259
Oxygen	0	316
Air	0	331
Air	20	343
Helium	0	965
Liquids		
Chloroform	20	1004
Ethanol	20	1162
Mercury	20	1450
Water	20	1482
Solids		
Lead	–	1960
Copper	–	5010
Glass	–	5640
Steel	–	5960

- Sound can be reflected by objects
- While propagating, sound loses energy attenuating the amplitude



REVERBERATION AND EVENT LOCALIZATION

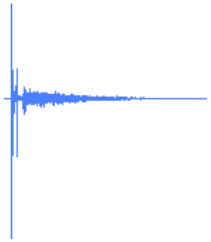




- Soundwave propagates towards all the directions
- Rays (soundwave) are partially reflected and partially absorbed by objects
- Multiple replicas of the same sound arrive to the listener with different delays: reverberation: defines the location of the event in the room
- Identify event and reverberation can also be a scene classification strategy



1st Baptist Church
Nashville

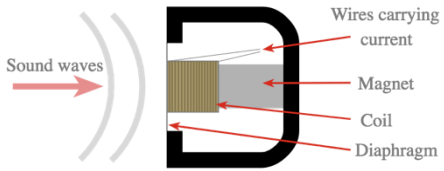


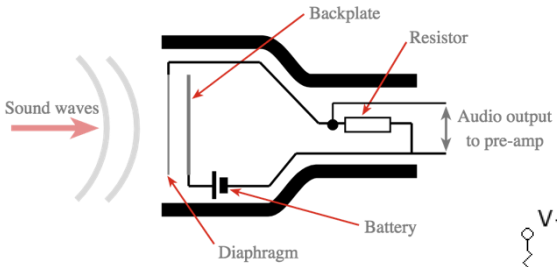


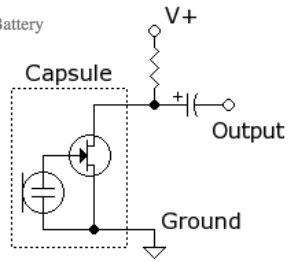
Troller's Gill
Yorkshire Dales



MICROPHONE'S STRUCTURE

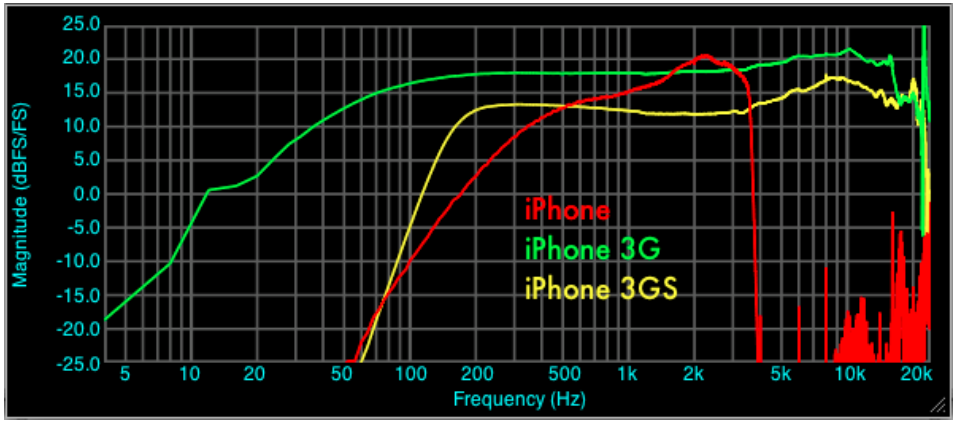


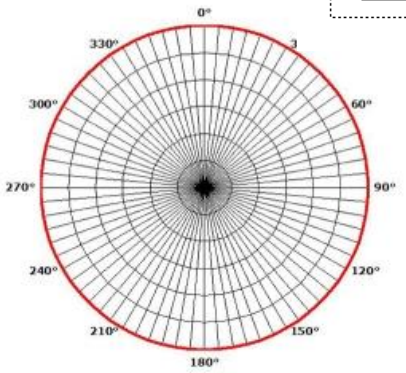




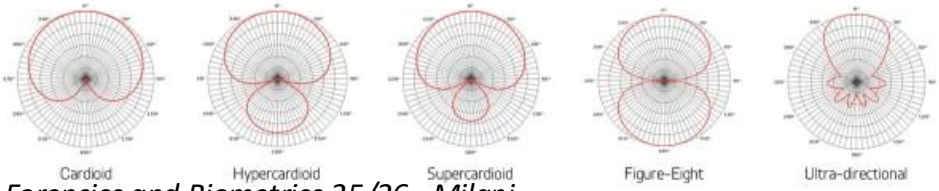
Characteristics:

- transducer type, { dynamic, condenser, electret condenser.
- sensitivity, { Electrical, mechanical
- frequency response,
- directionality (polar pattern)

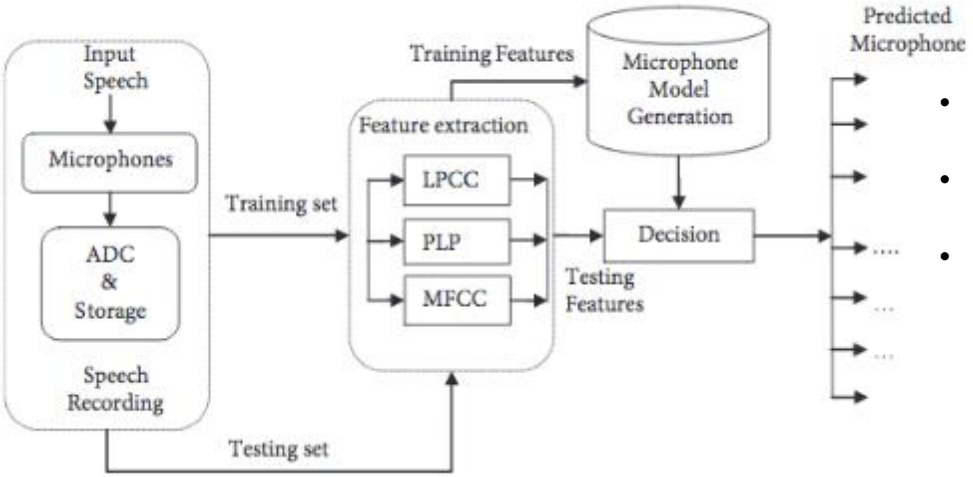




Omni-directional



FEATURE EXTRACTION



- LPCC: linear prediction cepstrum coefficients
- PLCPP: perceptually-based linear predictive coefficients
- MFCC: mel-frequency cepstral coefficients

